

Dashboard design

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD

President, Scrivner Solutions Inc

Why design is important?

- **Company:** NASDAQ
- **Needs:** communicate every day's market to the average investor
- **Solution:** a real-time single value display at Times Square

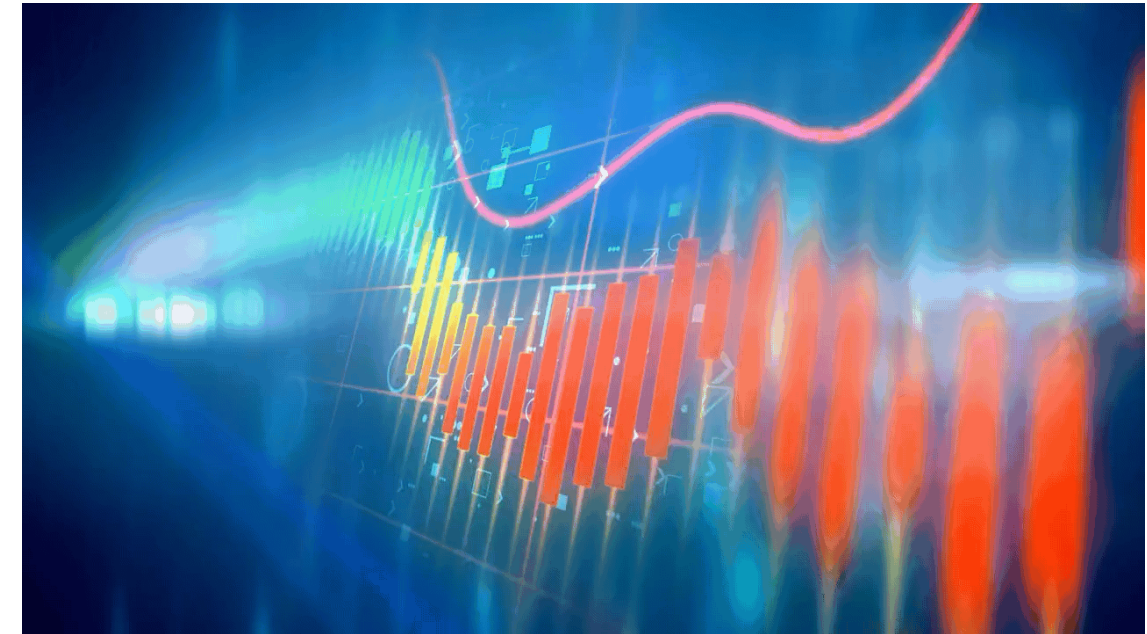


Why trend lines are not the best solution?

¹ Brath, Richard & Michael Peters. (2004). Dashboard design: why design is important.

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Why trend lines are not the best solution?

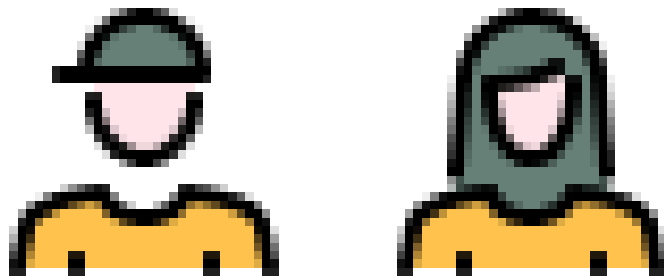
Historical insights versus real-time indicators

¹ Brath, Richard & Michael Peters. (2004). Dashboard design: why design is important.

Design components

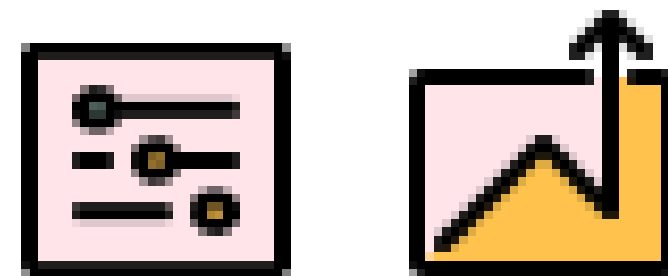
Audience

- Who will be using this dashboard?
- What are their needs?
- What format is suitable for the audience?
- How often will it be accessed?



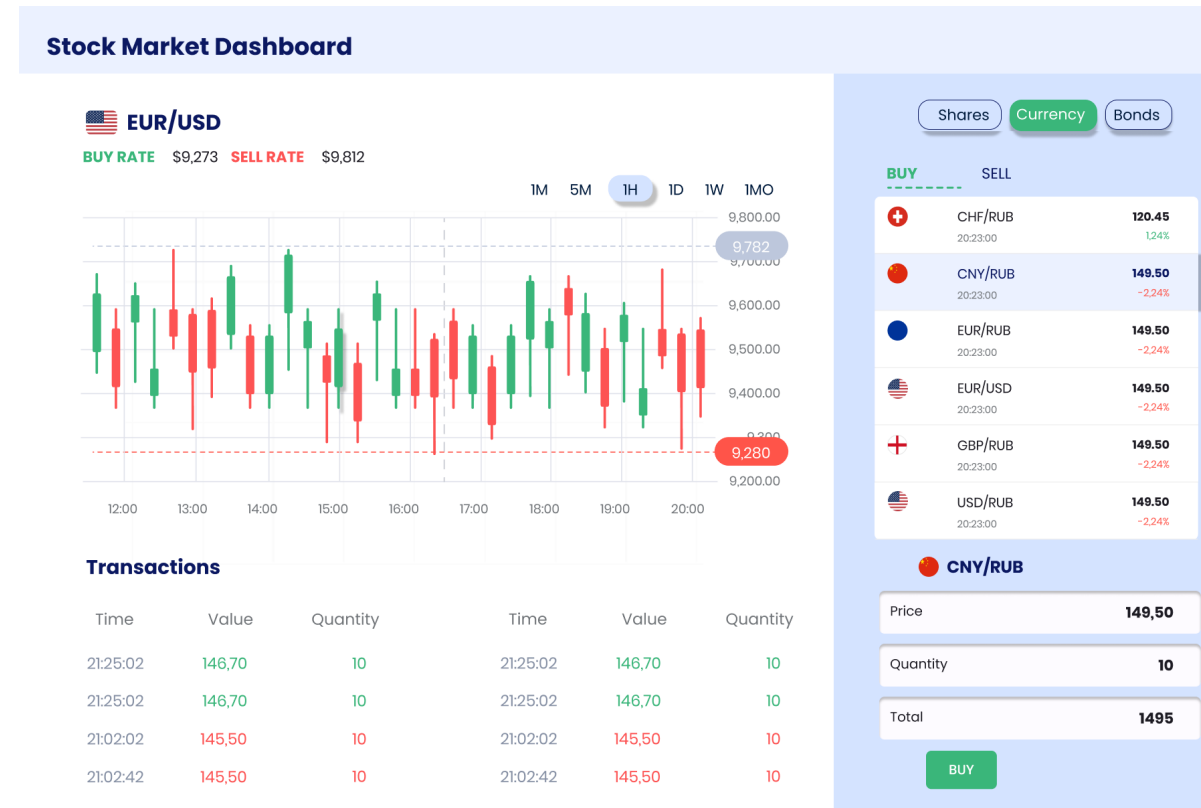
Components

- **Purpose:** exploratory versus explanatory
- **Format:** horizontal versus vertical
- **Layout:** grid versus flex
- **Visuals:** values, dimensions, shapes



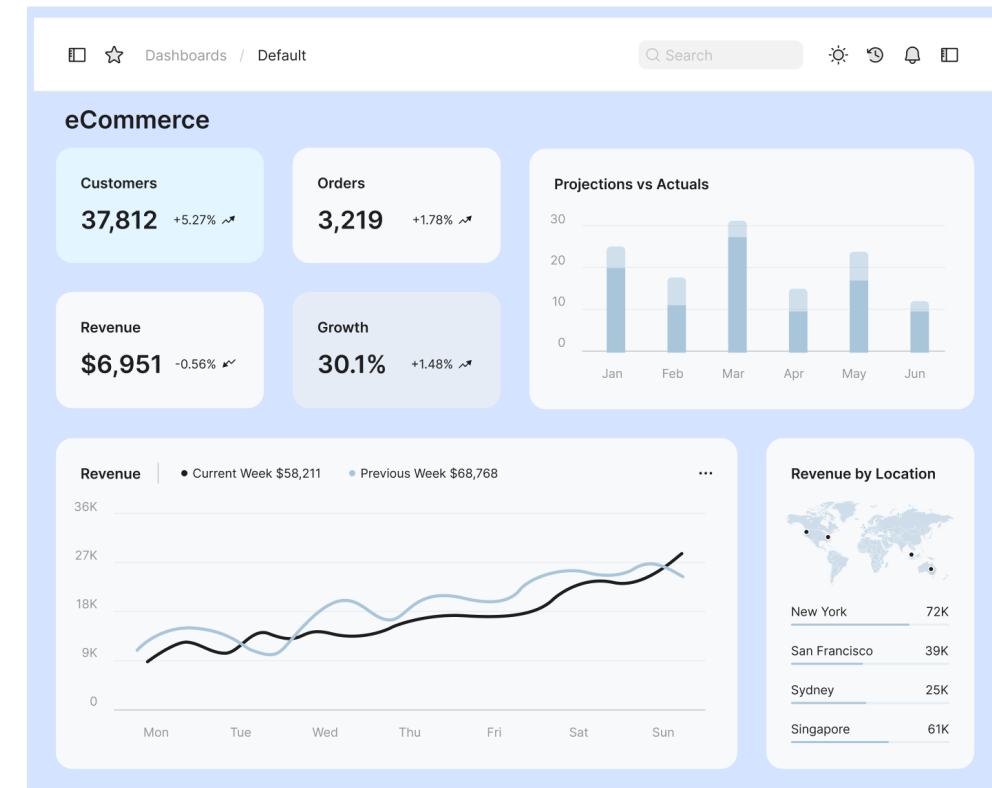
Dashboard purpose

Exploratory dashboard



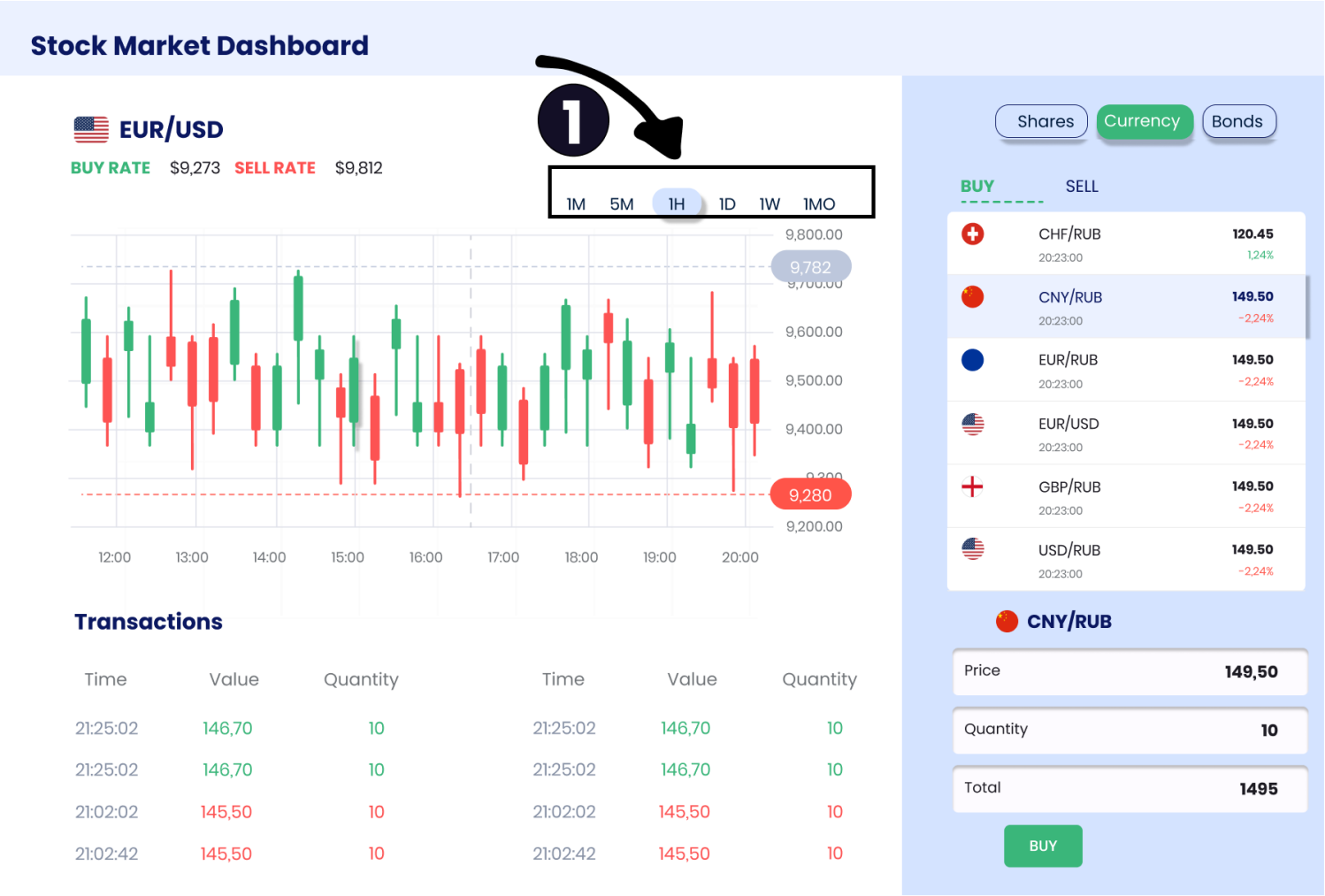
An exploratory dashboard allows users to explore data.

Explanatory dashboard



An explanatory dashboard provides users with predefined insights.

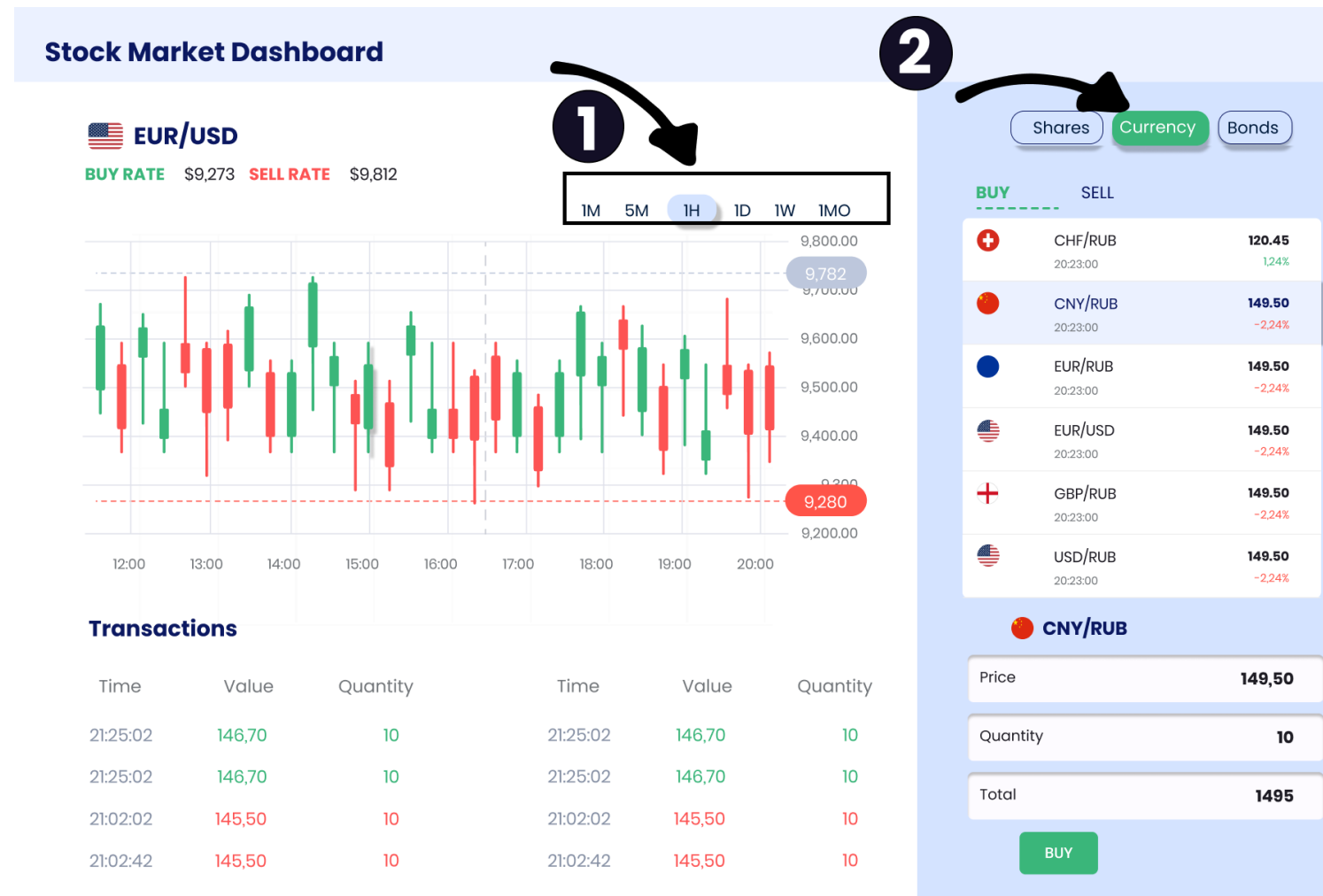
Exploratory dashboard



- Filters for custom data segmentation

¹ Baucke, Spencer. (2022). Dashboard design essentials.

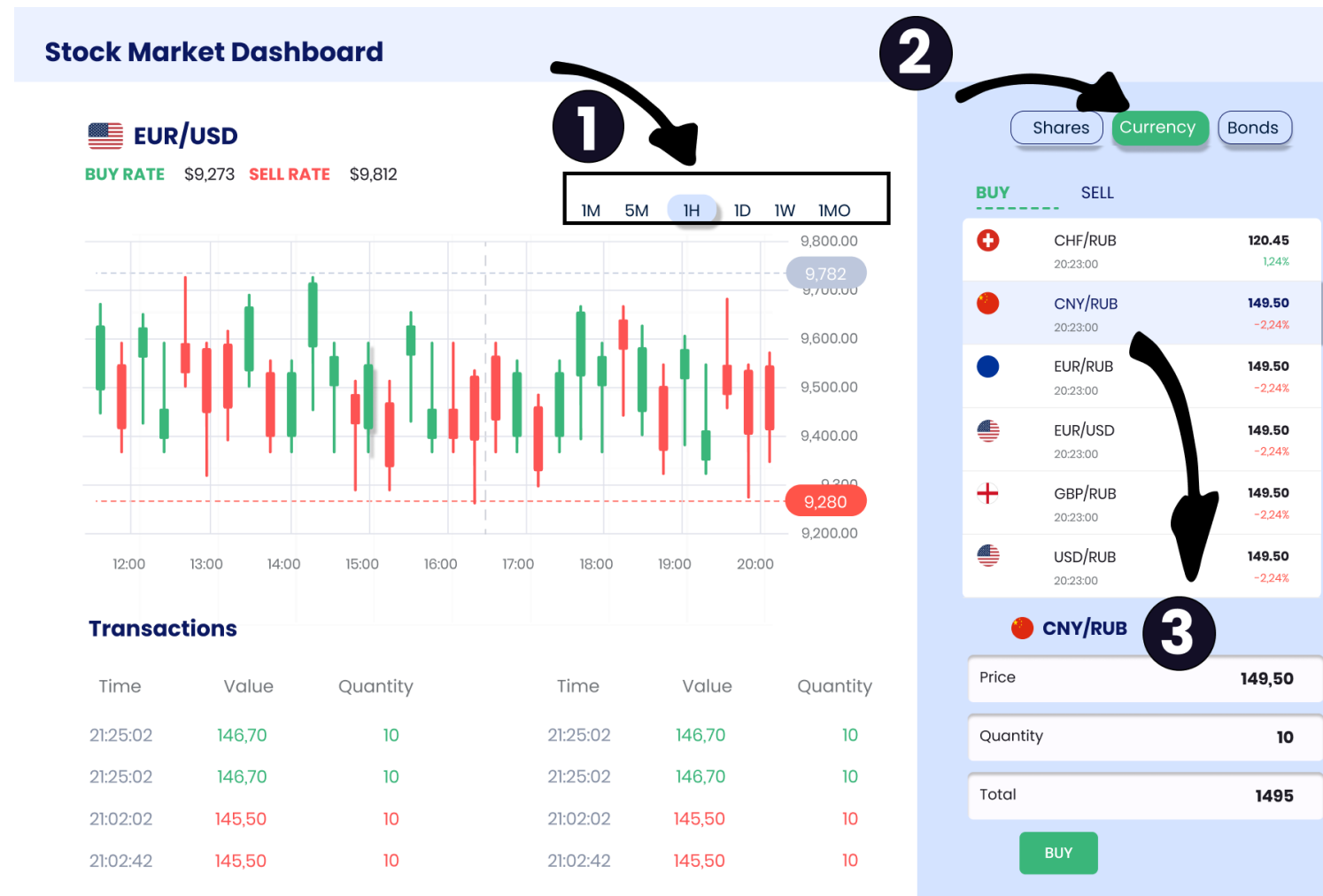
Exploratory dashboard



- Filters for custom data segmentation
- Customizable navigation panels

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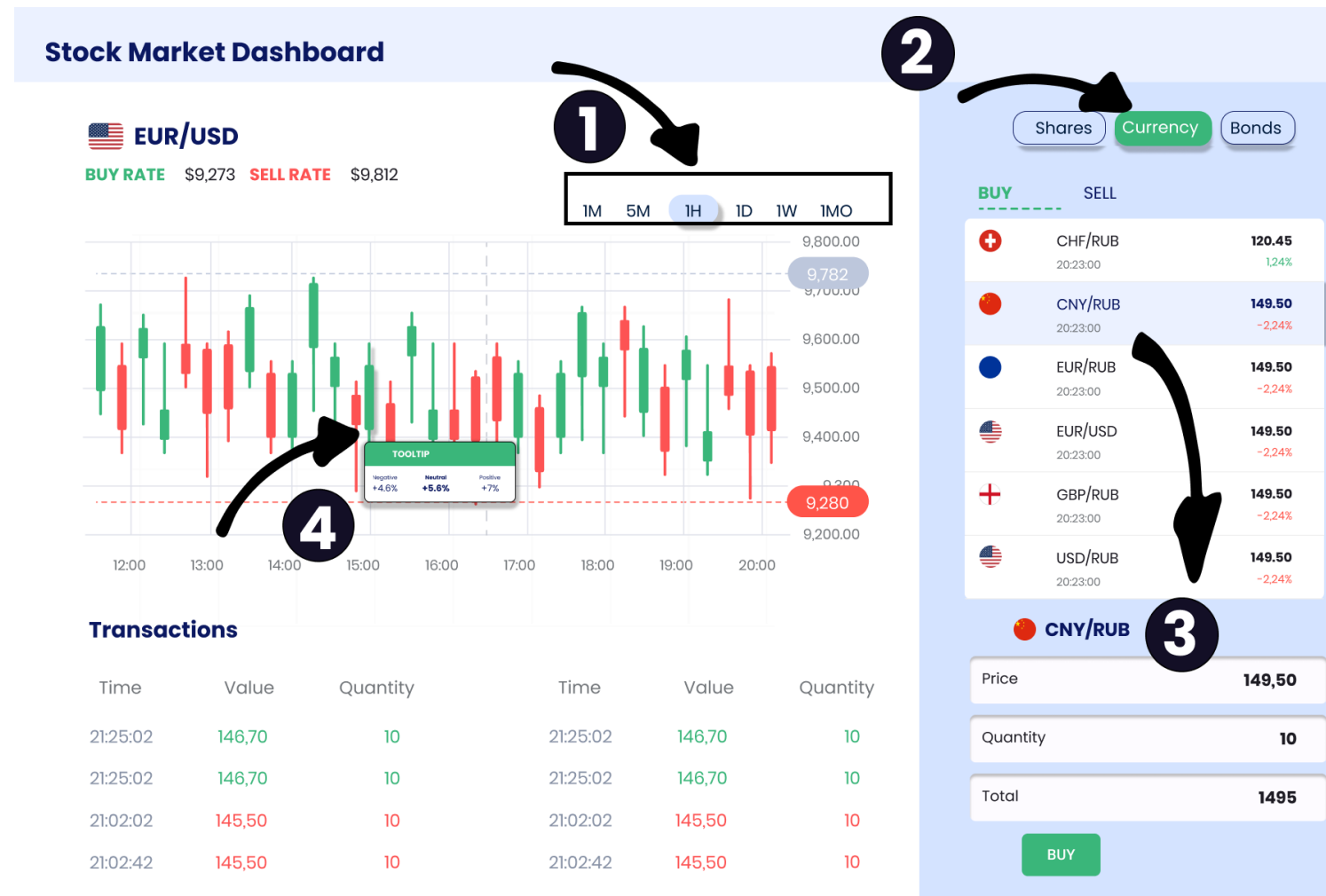
Exploratory dashboard



- Filters for custom data segmentation
- Customizable navigation panels
- Drill-down for more granular view

¹ Baucke, Spencer. (2022). Dashboard design essentials.

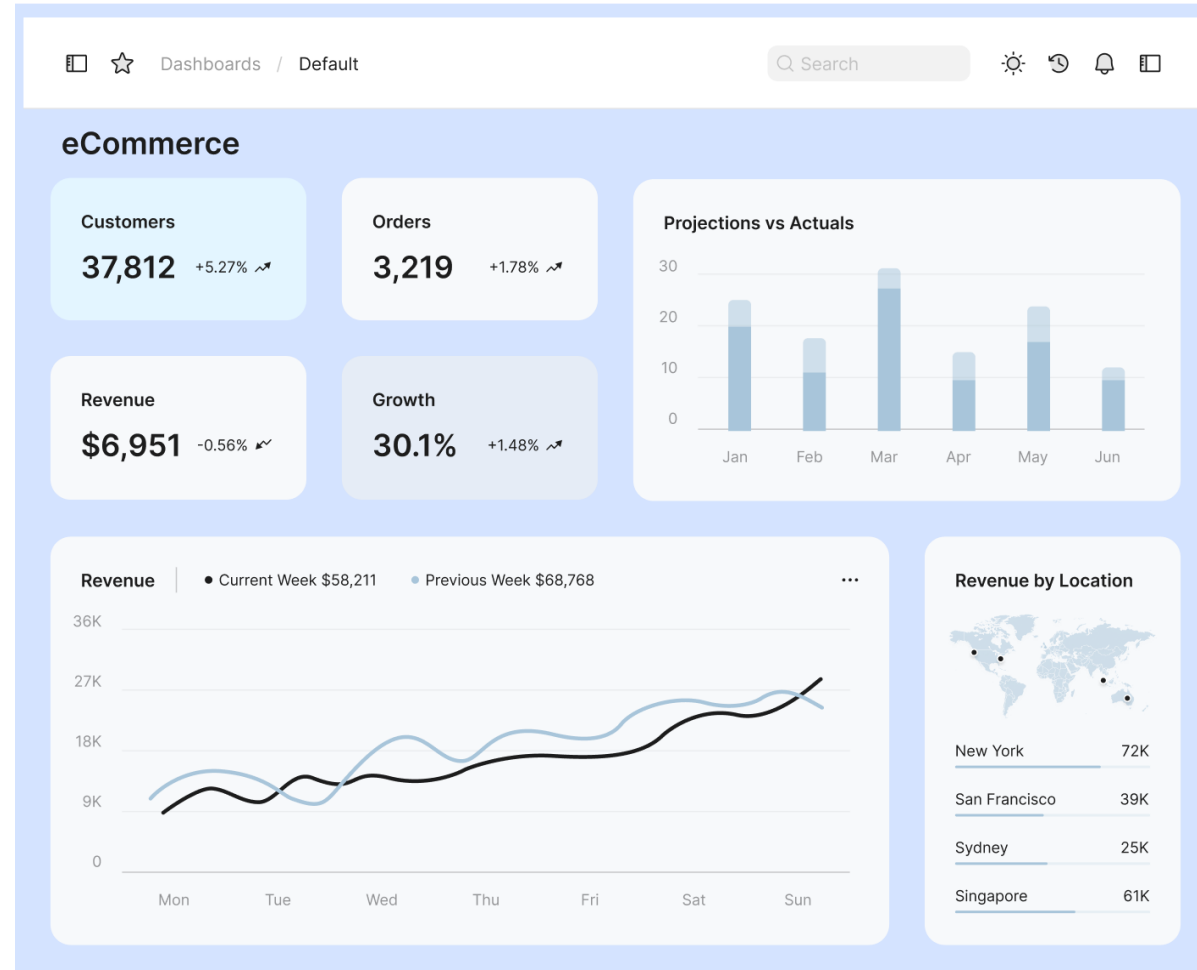
Exploratory dashboard



- Filters for custom data segmentation
- Customizable navigation panels
- Drill-down for more granular view
- Interactive visuals (hover-over, tooltips)

¹ Baucke, Spencer. (2022). Dashboard design essentials.

Explanatory dashboard



- Data is often already filtered
- Predefined storyline
- Presentation to executives
- A call for action

¹ Baucke, Spencer. (2022). Dashboard design essentials.

Dashboard size

Physical space

- **Monitor size** - a diagonal screen distance

Desktop, laptop, tablet, mobile devices



Dashboard size

Physical space

- **Monitor size** - a diagonal screen distance
- **Screen resolution** - the number of pixels and their density (width x height)

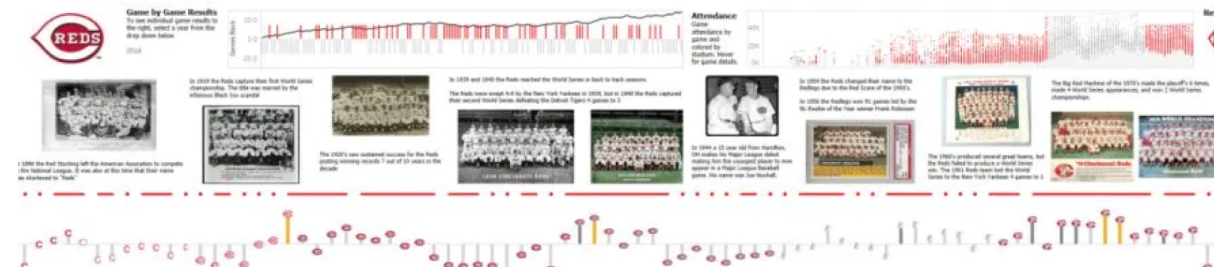
Format

- **Standard format** - a rectangular dashboard
 - Ratio - 16:9 ratio
 - All components visible on a screen
 - No scrolling
- **Long format** - horizontal or vertical

Desktop, laptop, tablet, mobile devices



Horizontal (timeline style)



Dashboard size

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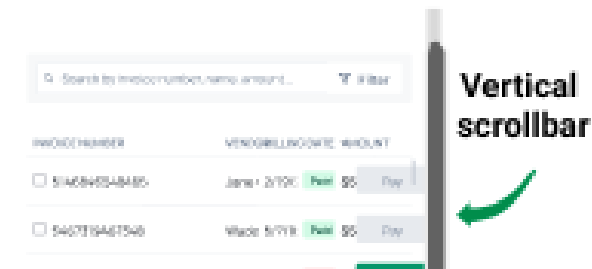
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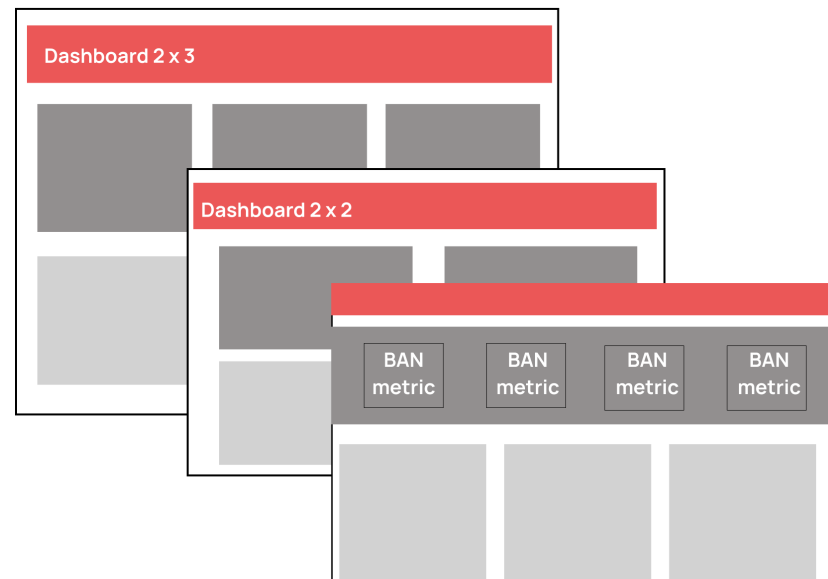


Vertical (newspaper column style)



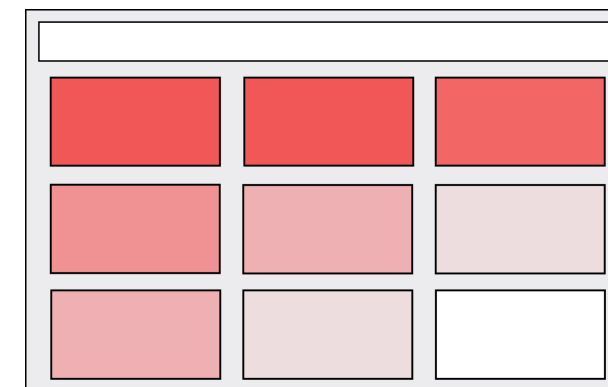
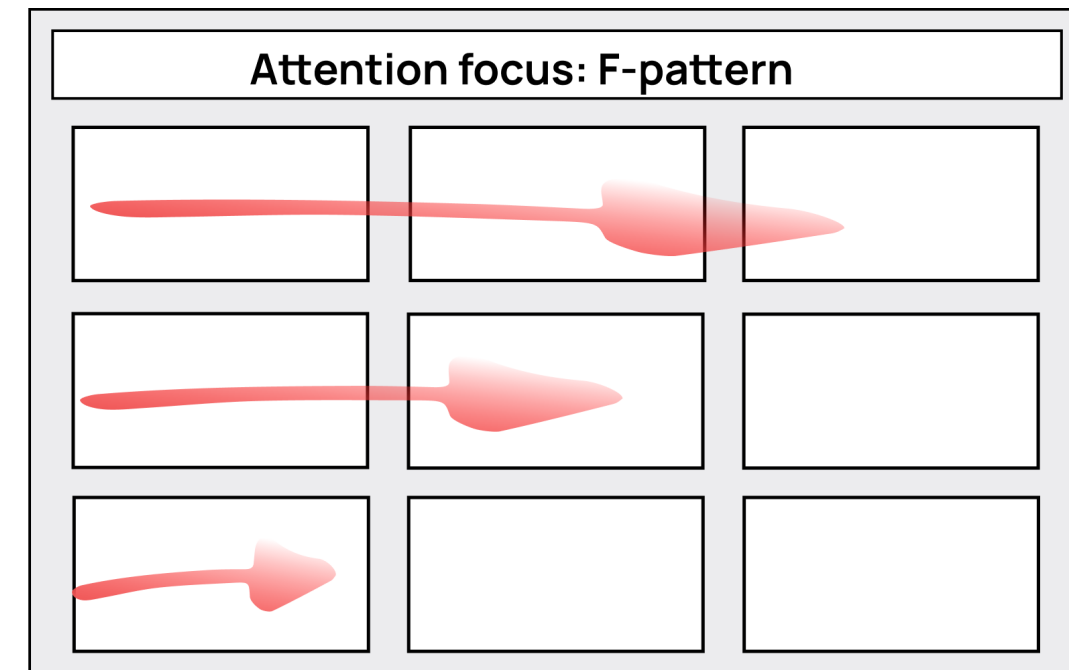
Dashboard layout

Grid system



- Two-dimensional matrix
- Balanced and symmetrical
- Simple navigation
- Big Aggregate Numbers (BAN) - top row

Focus



¹ Baucke, Spencer. (2022). Dashboard design essentials.

Let's practice!
DASHBOARD DESIGN CONCEPTS

Dashboard-specific visualization techniques

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD
President, Scrivner Solutions Inc

Designing dashboards for human eyes

Effective data visualization

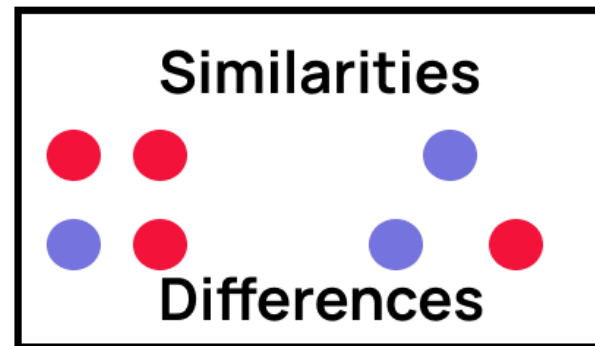
- Reduces cognitive load on the user
- Aids in problem solving
- Facilitates discovery
- Provides actionable insights

well designed data graphics are usually the simplest and [...] the most powerful¹

¹ Edward Tufte

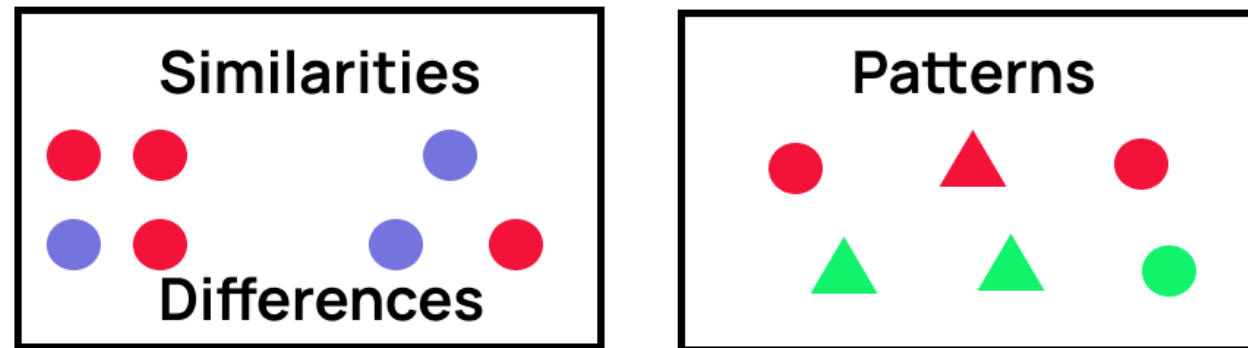
Design principles

Processing information



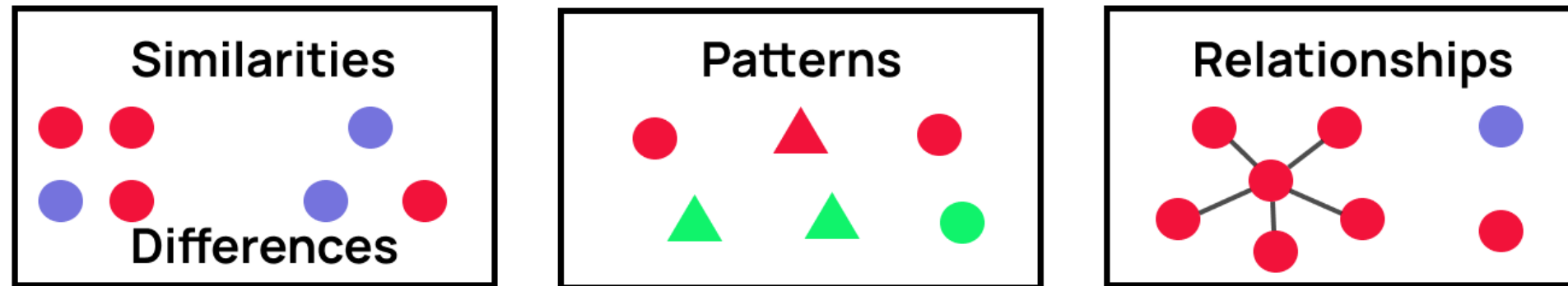
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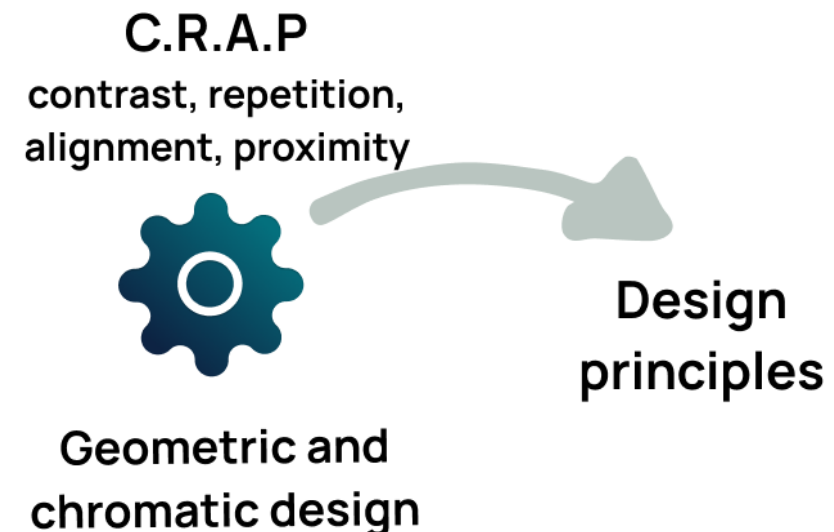


Design principles

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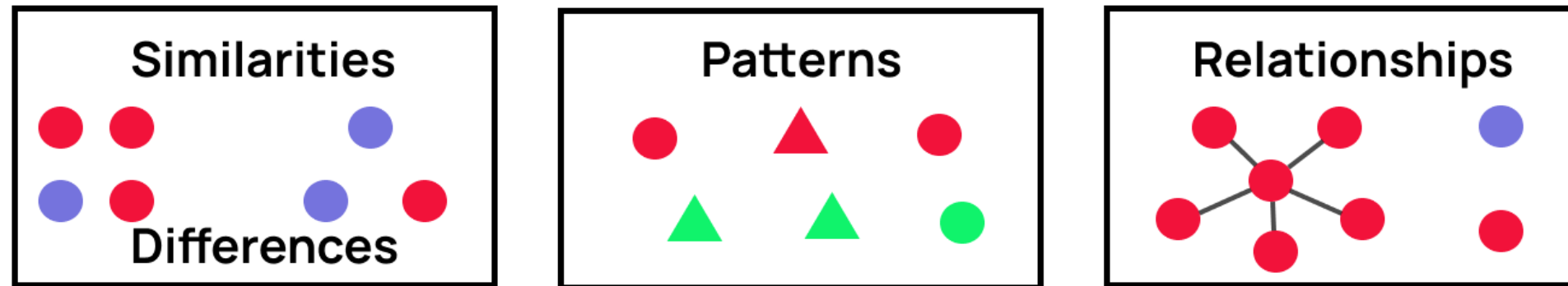


Dashboard design principles

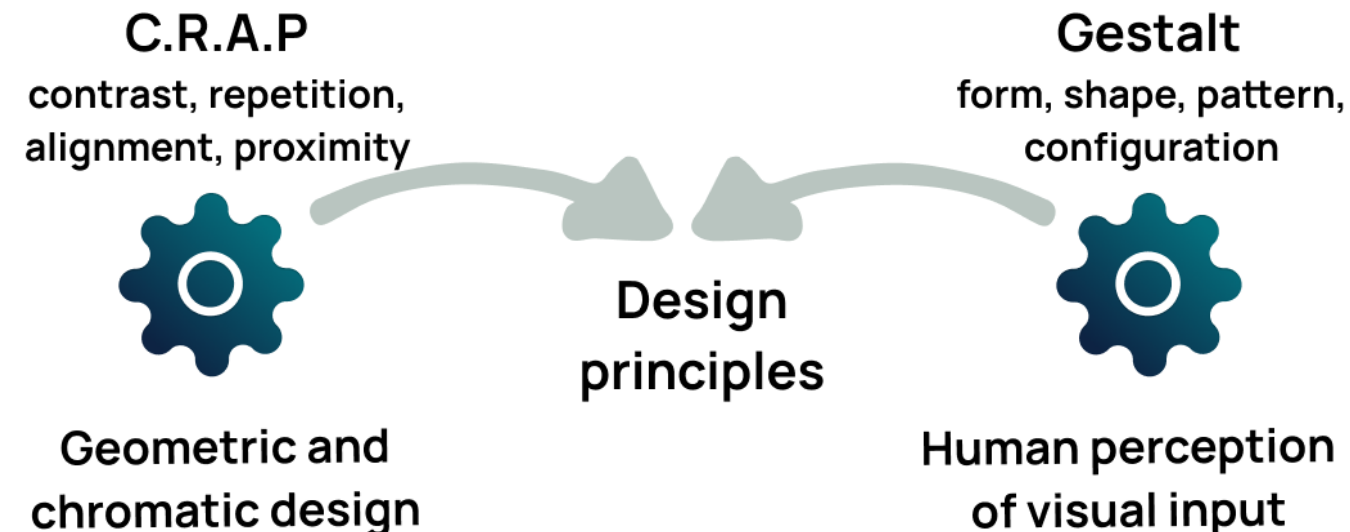


Design principles

Processing information



Dashboard design principles



C.R.A.P. design principles

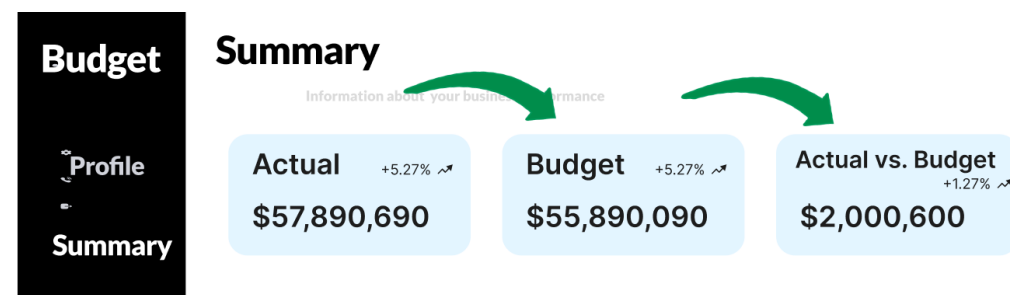
Contrast

- Color, size, shape

Social Media Campaign	KPI 1	KPI 2	KPI 3
Impressions	526	829	1,231
Growth Last 7 days	-	58.21%	-31.31%
Accounts Reached	112	253	422
Growth Last 7 days	-	125.89%	-35.35%

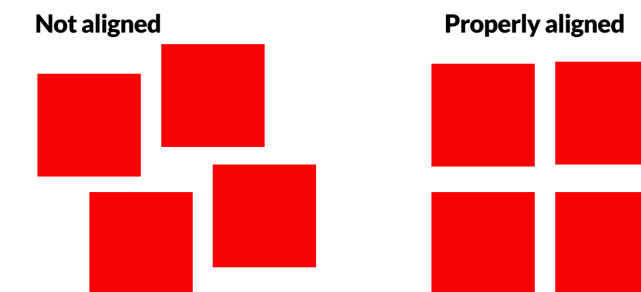
Repetition

- The "rhythm" of the dashboard reading



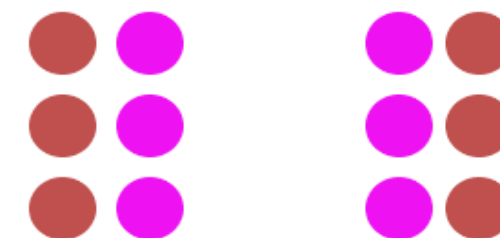
Alignment

- An ordered design



Proximity

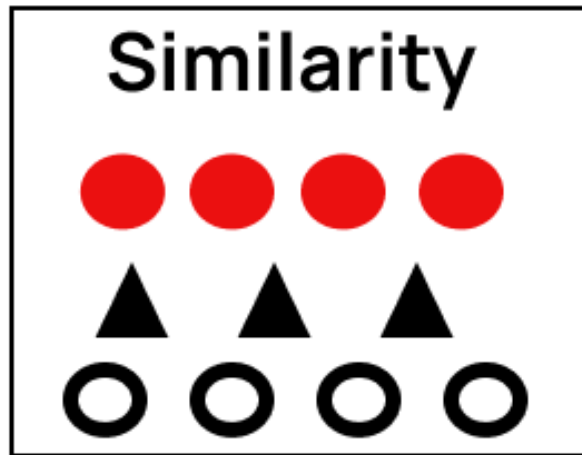
- Related elements must be grouped



¹ Ferrari, Maw. (2021). Improving Dashboard design.

Gestalt principles

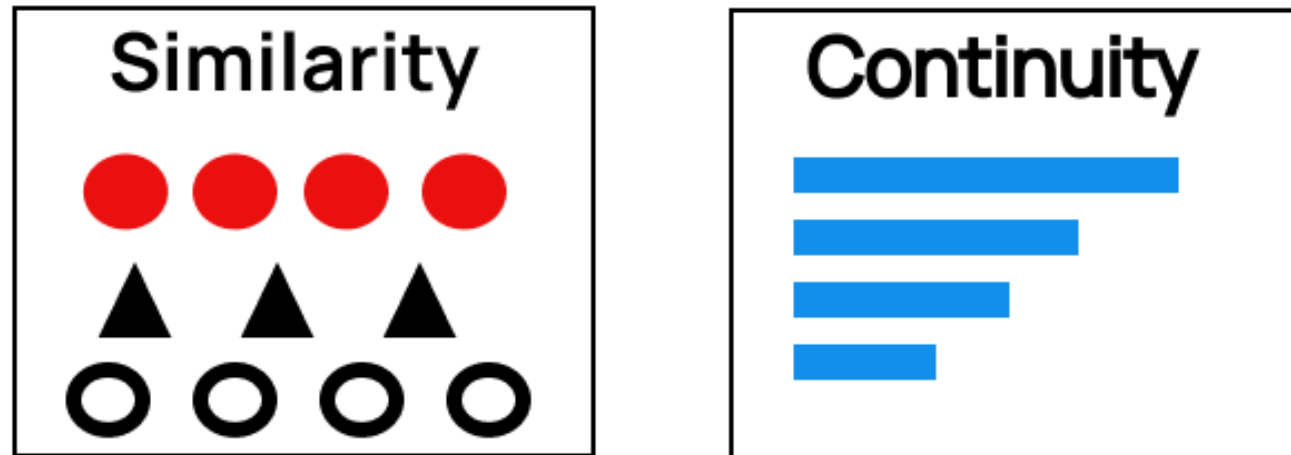
How we perceive and organize what we see in particular ways in an effort to make sense of it.



¹ Dataspire. (2020). Let's leverage perception science to our advantage!

Gestalt principles

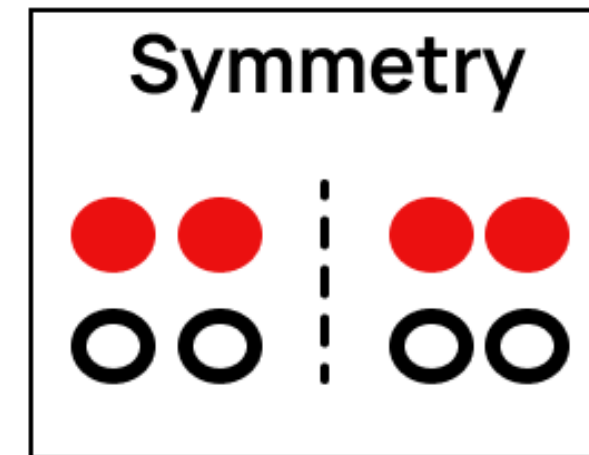
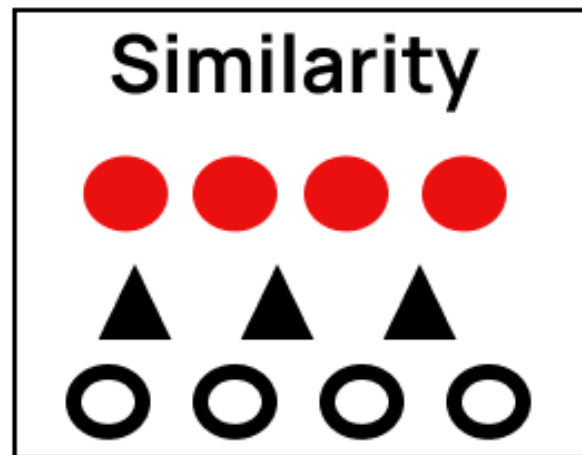
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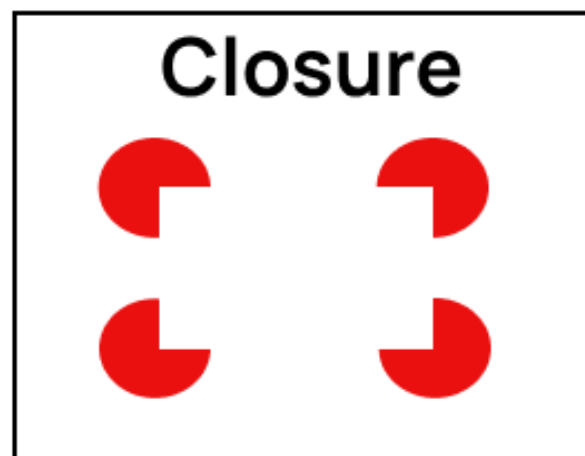
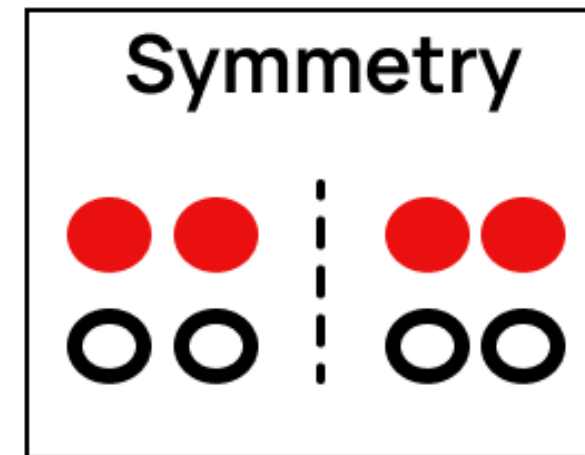
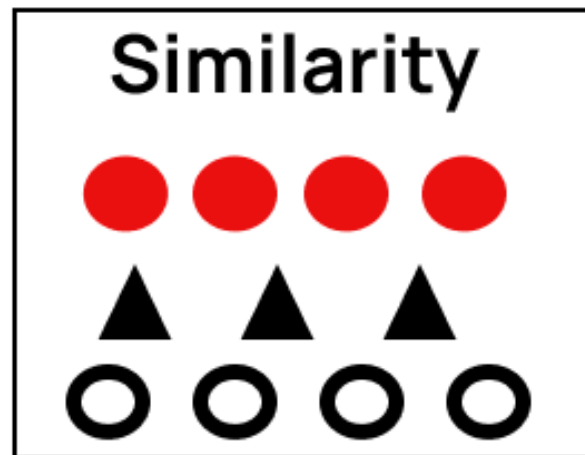
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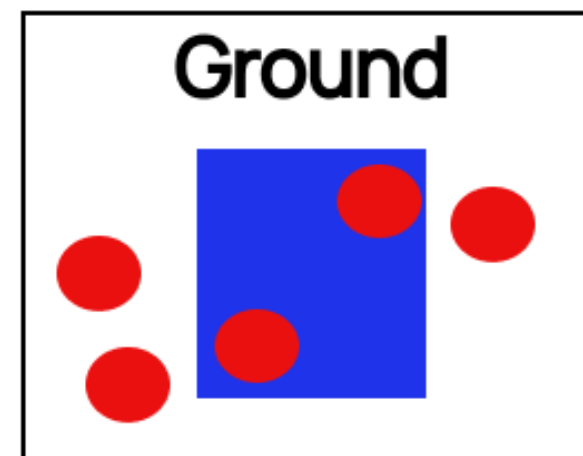
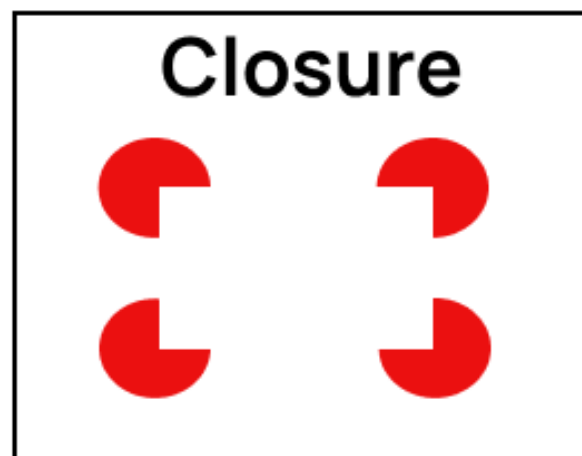
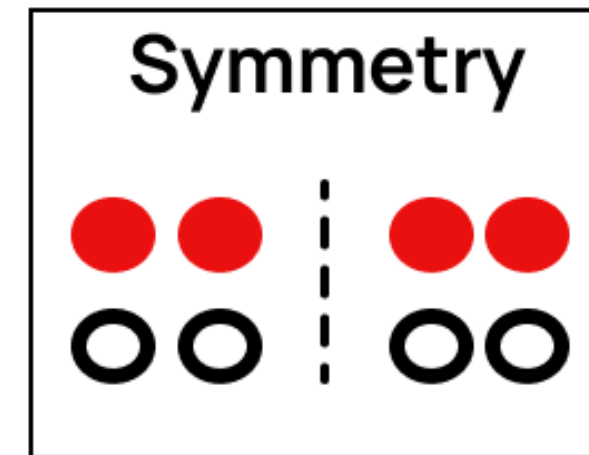
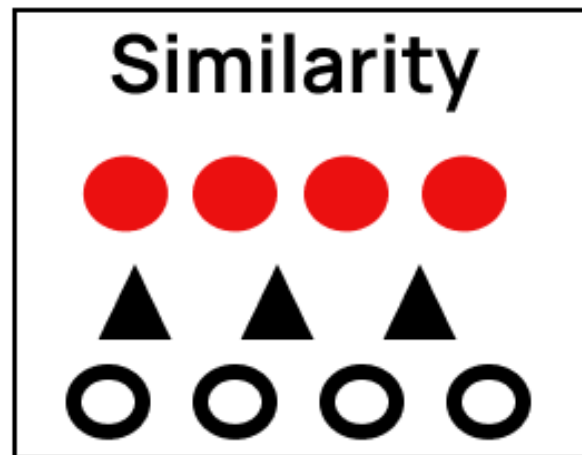
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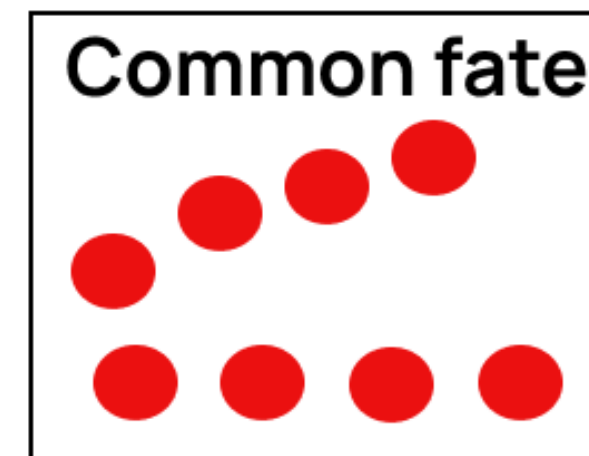
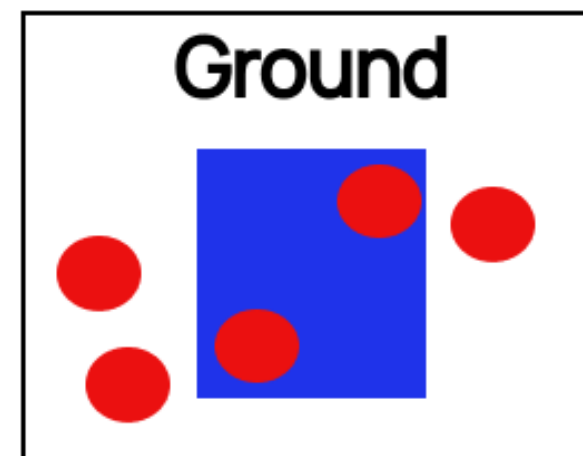
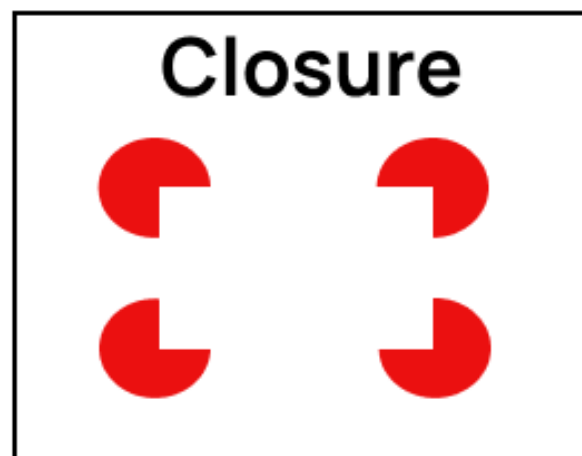
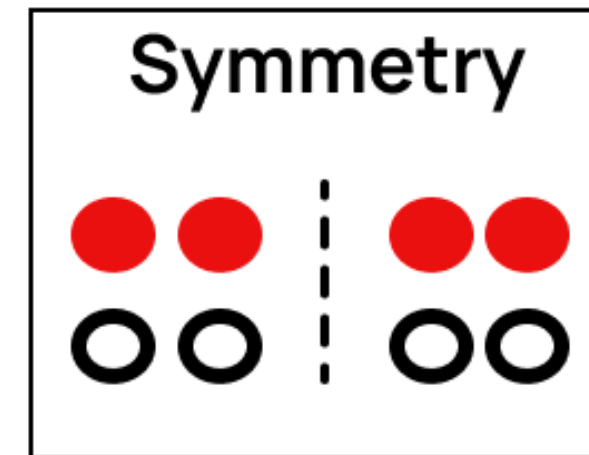
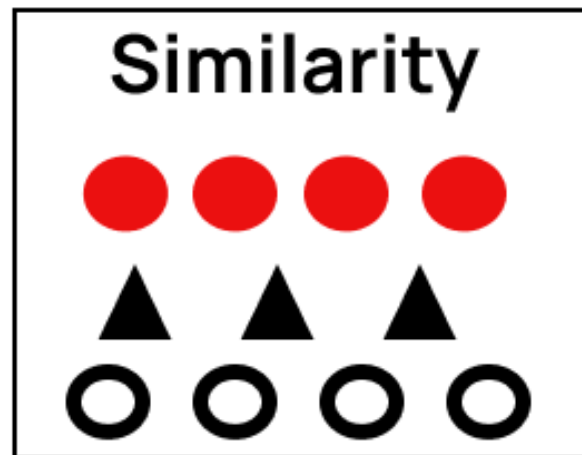
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Gestalt principles

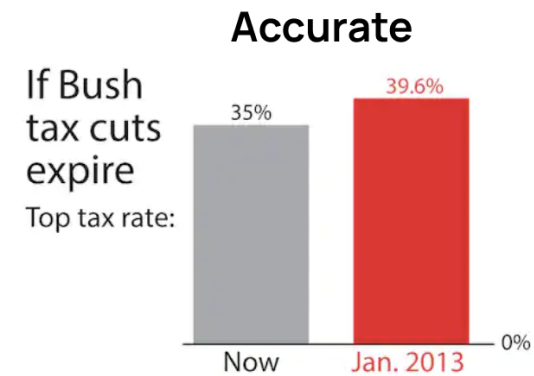
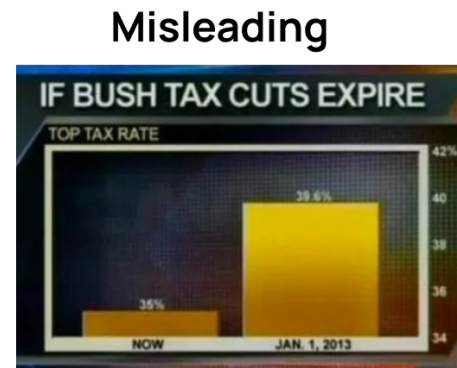
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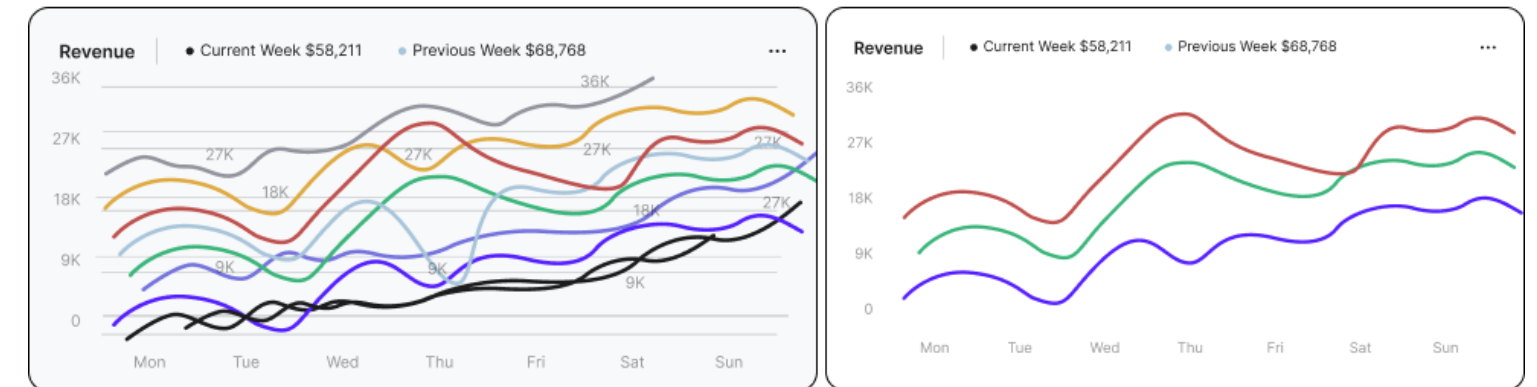
¹ Dataspire. (2020). Let's leverage perception science to our advantage!

Misleading visualization: dos and don'ts

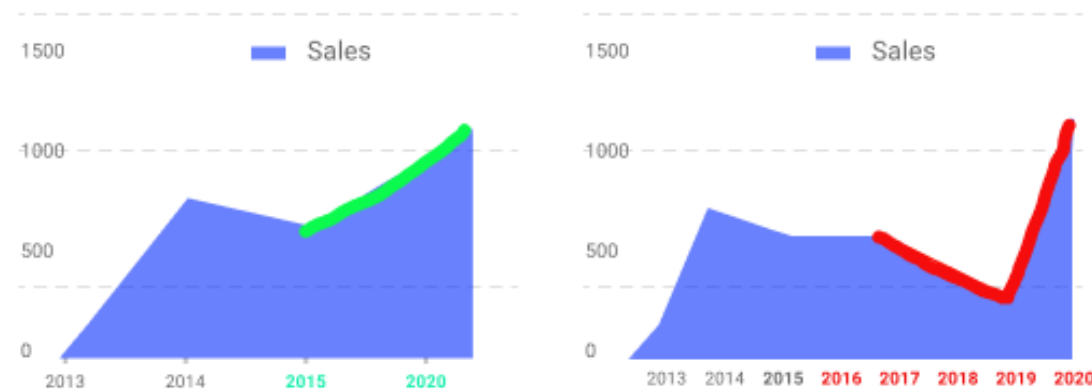
Non-zero baseline



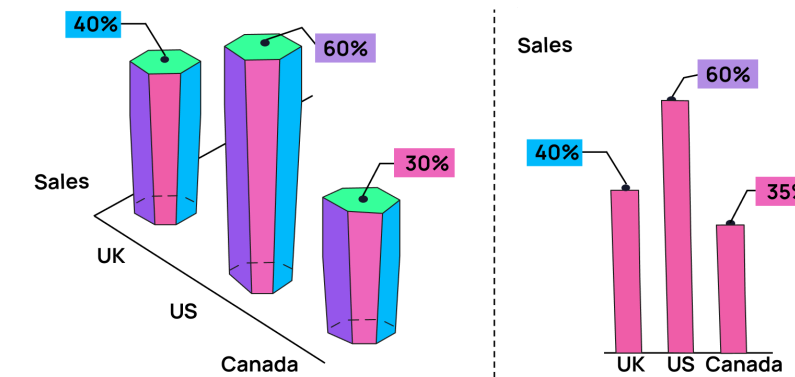
Overload



Interval inconsistency



3D effects



¹ LibGuide. (2021). Data Visualization: Chart Dos and Don'ts.

Let's practice!
DASHBOARD DESIGN CONCEPTS

Principles of visual analytics

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD
President, Scrivner Solutions Inc

Visual analytics

Information design

Methods to make data meaningful

Data visualization

Visual methods of displaying data

Evidence

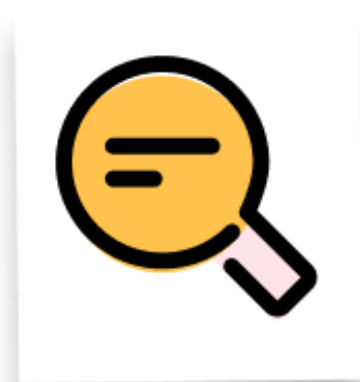


Visual analytics

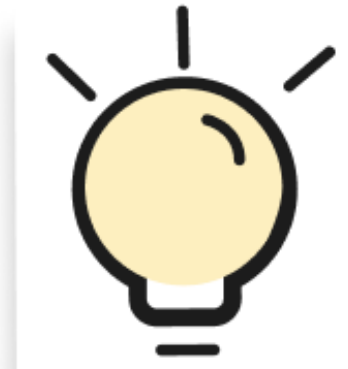
Information design

Methods to make data meaningful

Evidence



Reasoning



Data visualization

Visual methods of displaying data

Visual analytics

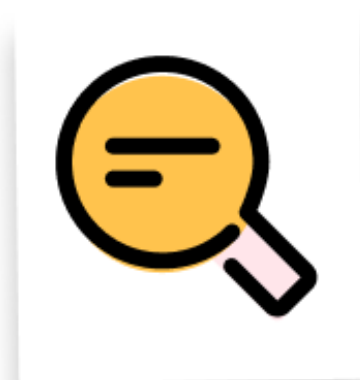
Information design

Methods to make data meaningful

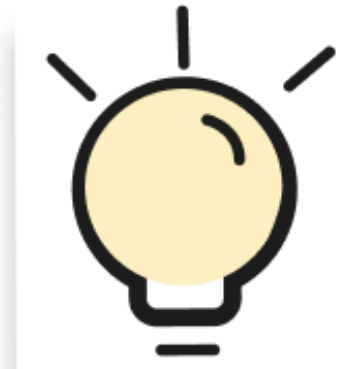
Data visualization

Visual methods of displaying data

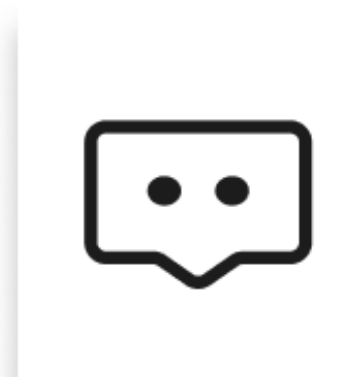
Evidence



Reasoning



Discourse



Fundamental principles of visual analytics

Principles of visual analytics are universal (Edward Tufte 2006)

¹ Thomas, James & Cook, Kristin. (2006). A visual analytics agenda.

Fundamental principles of visual analytics

Principles of visual analytics are universal (Edward Tufte 2006)

Meaningful
comparison

Causal
structure

Multivariate
links

Integration

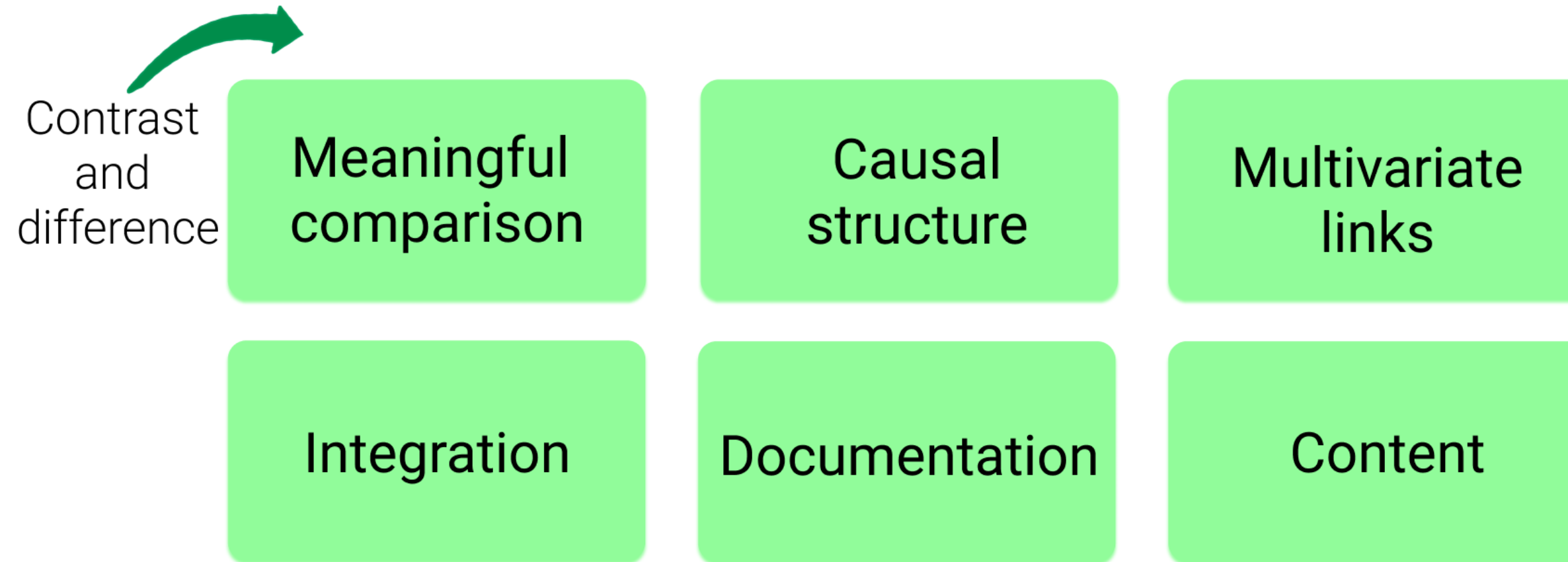
Documentation

Content

¹ Thomas, James & Cook, Kristin. (2006). A visual analytics agenda.

Fundamental principles of visual analytics

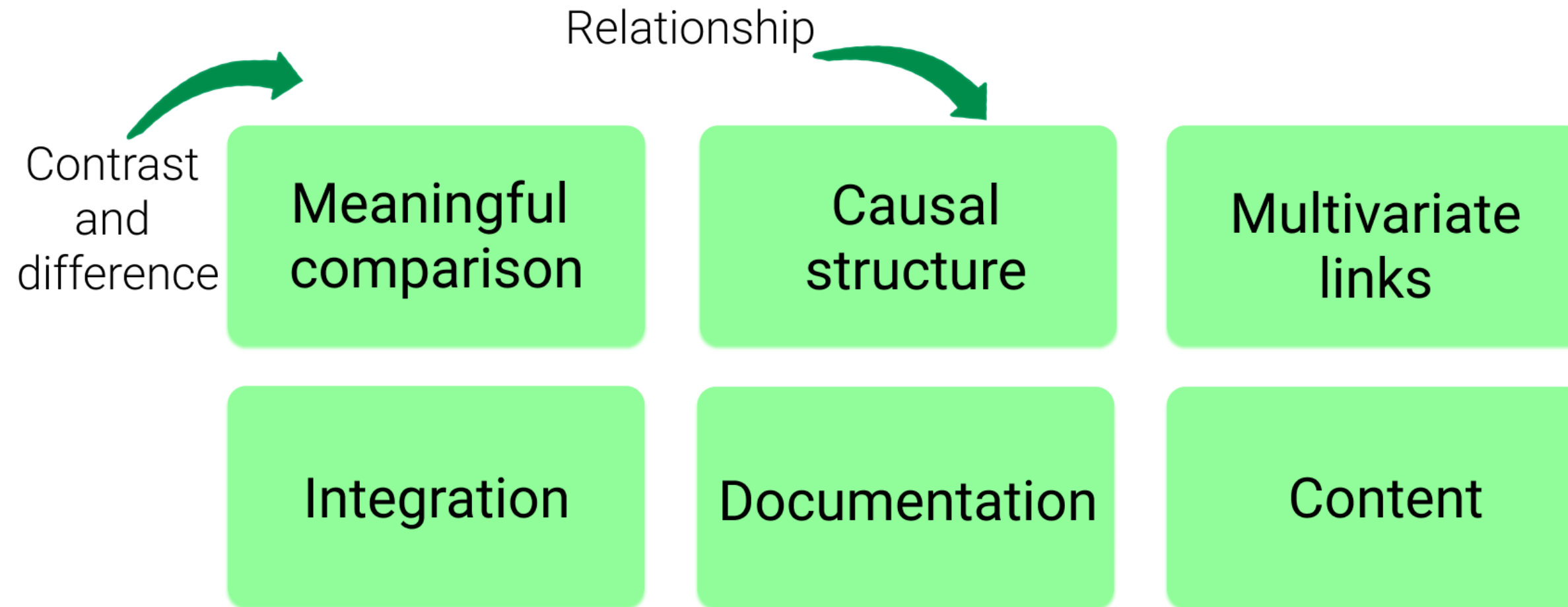
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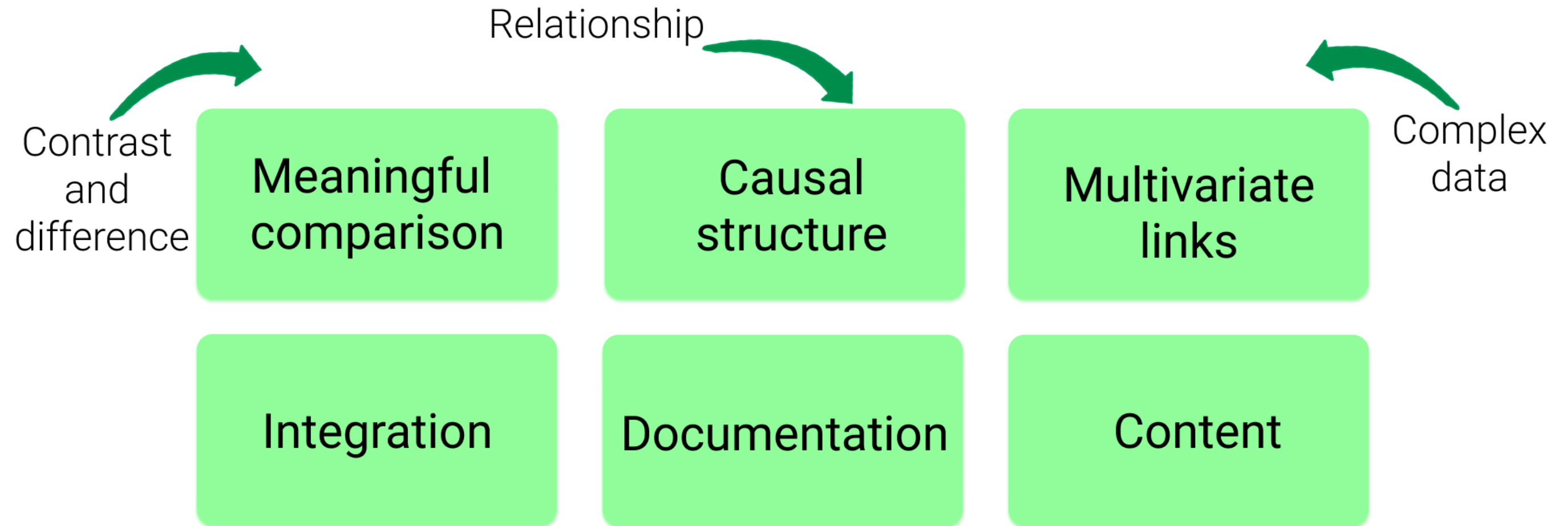
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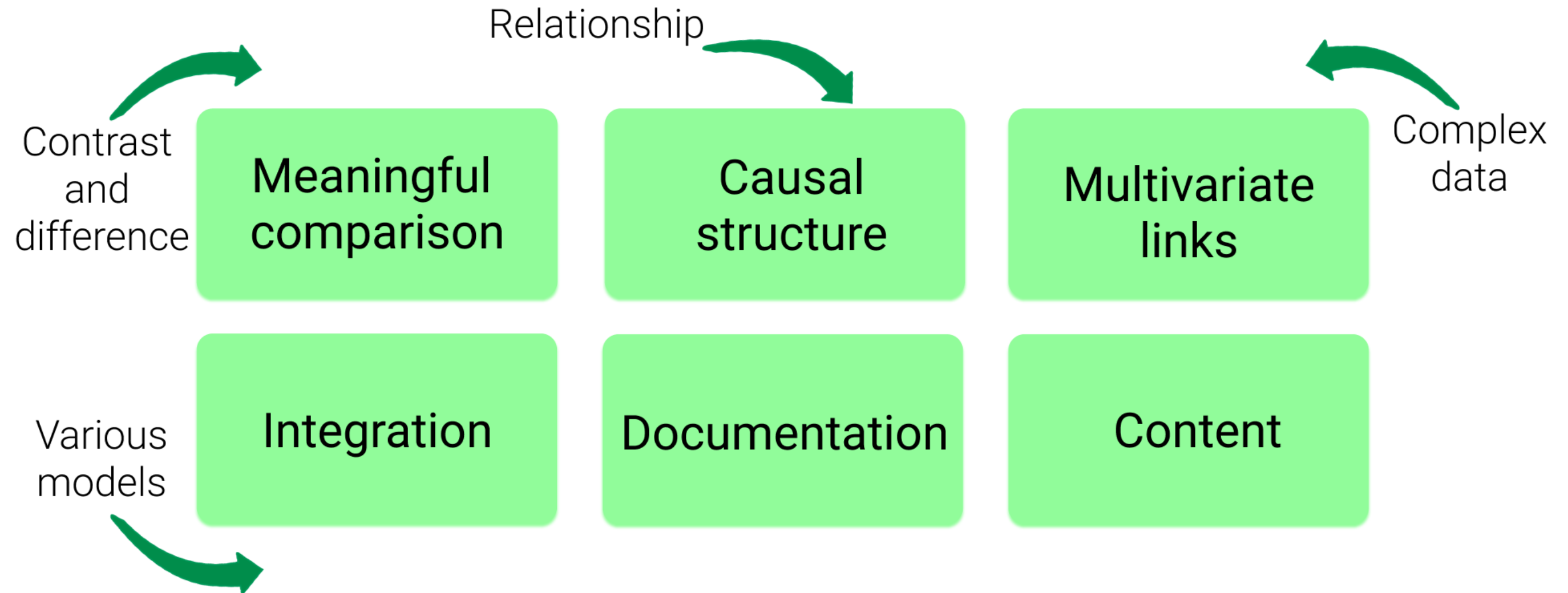
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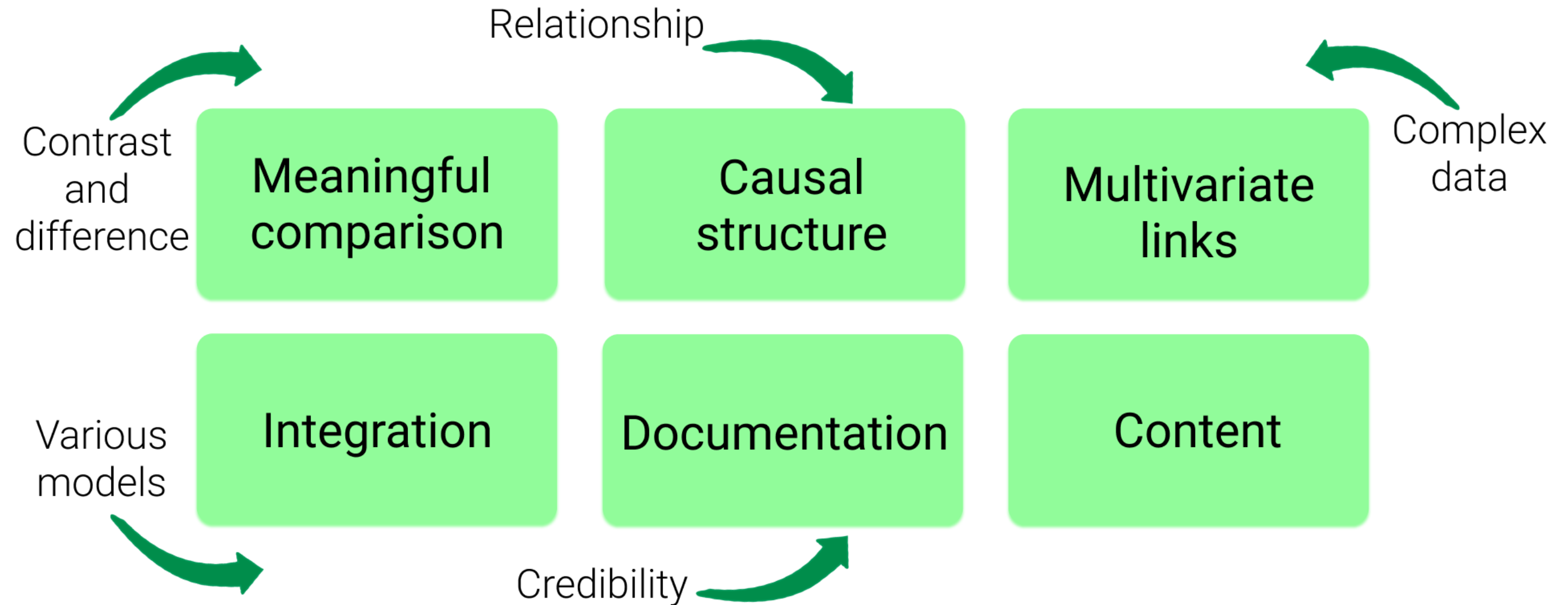
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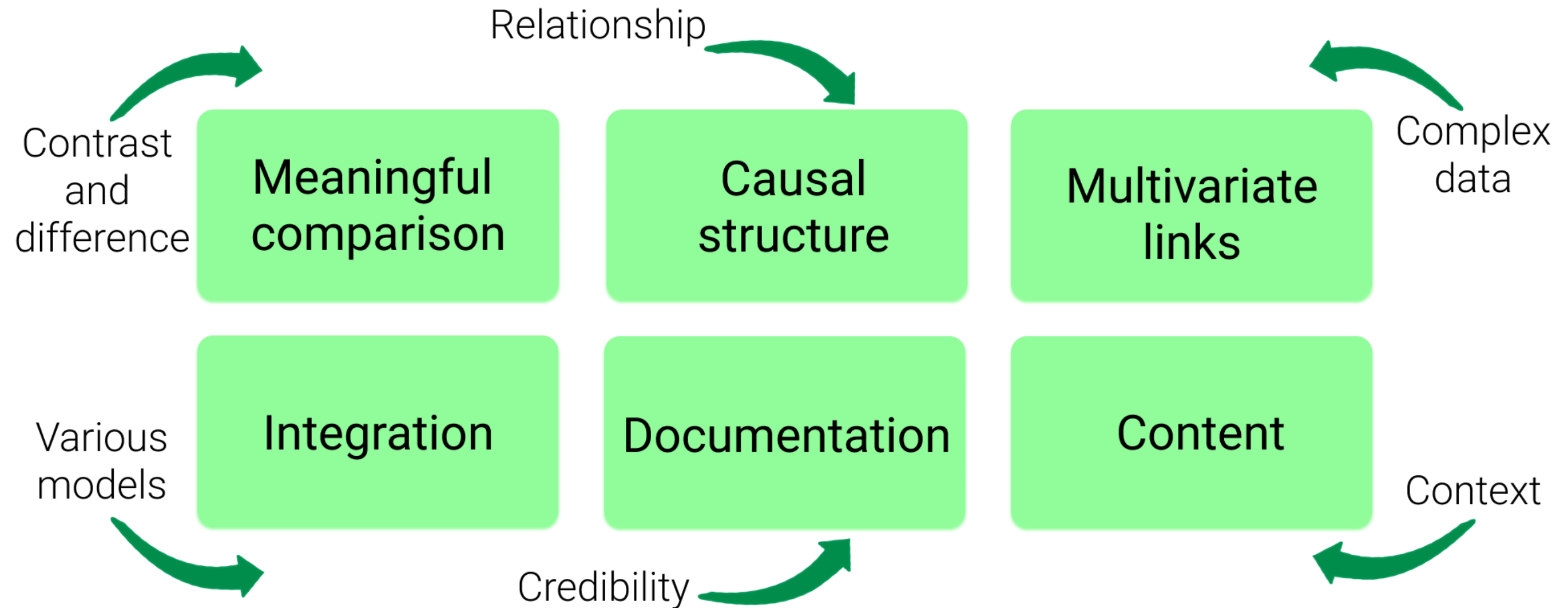
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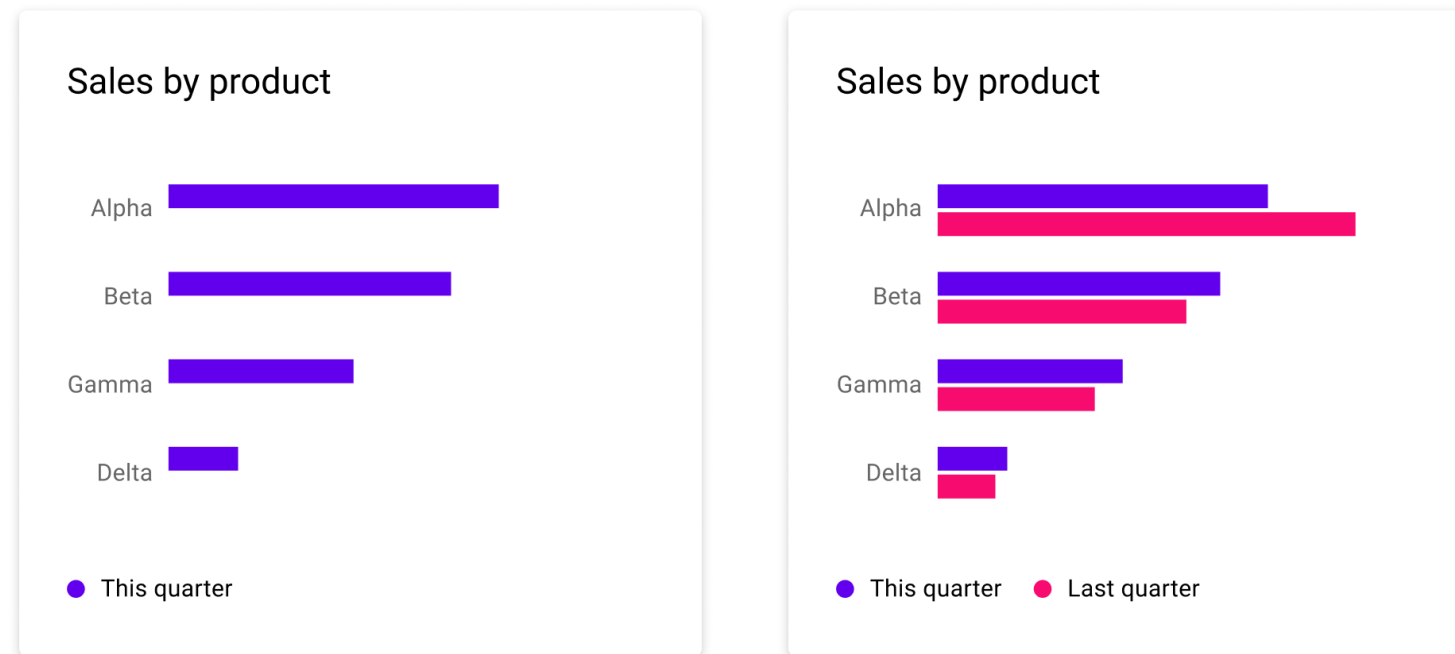


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Comparison and contrast

Show comparisons, contrasts, differences

"Compared to what?"



Not every comparison is meaningful

Time comparison

Comparison of **different components**

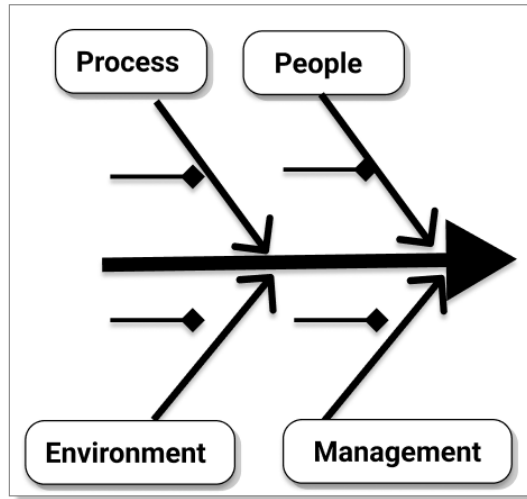
Comparison between **different groups**

¹ Edward Tufte (2006). Beautiful Evidence.

Causality and relationship

Show causality, mechanism, explanation, systematic structure

Diagram



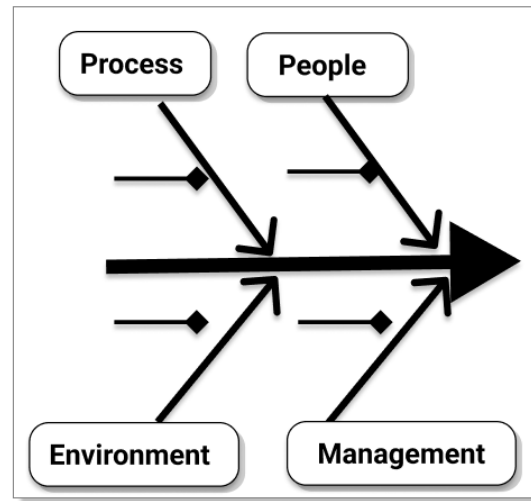
Sorting
ideas

¹ Edward Tufte (2006). Beautiful Evidence.

Causality and relationship

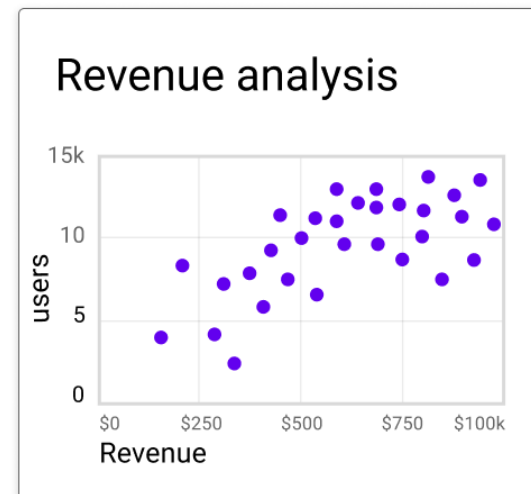
Show causality, mechanism, explanation, systematic structure

Diagram



Sorting
ideas

Scatterplot



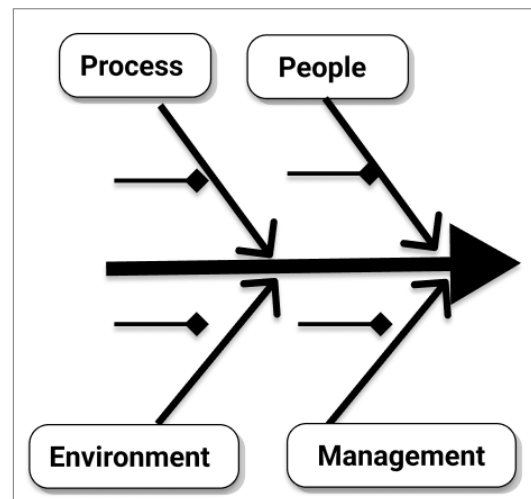
Simple
relationships

¹ Edward Tufte (2006). Beautiful Evidence.

Causality and relationship

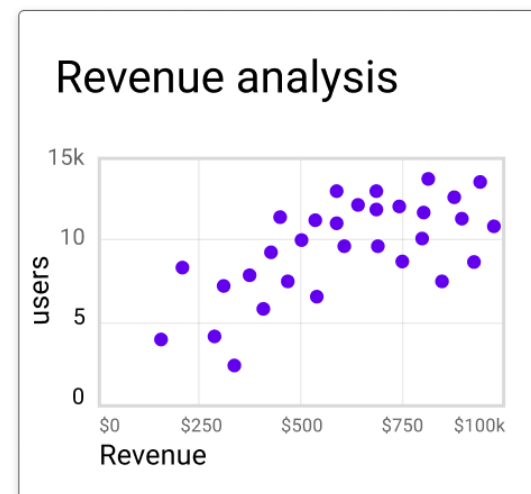
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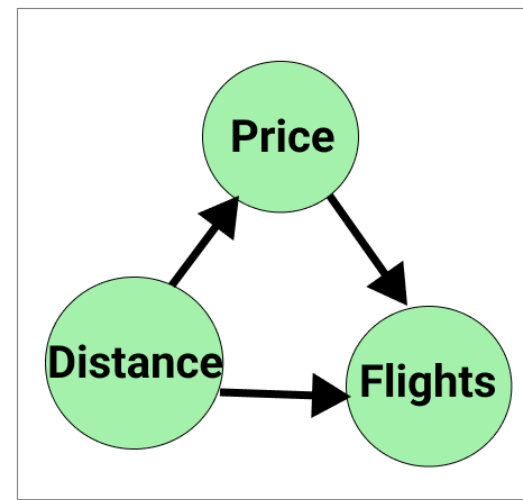
Sorting
ideas

Scatterplot



Simple
relationships

Causal
graph



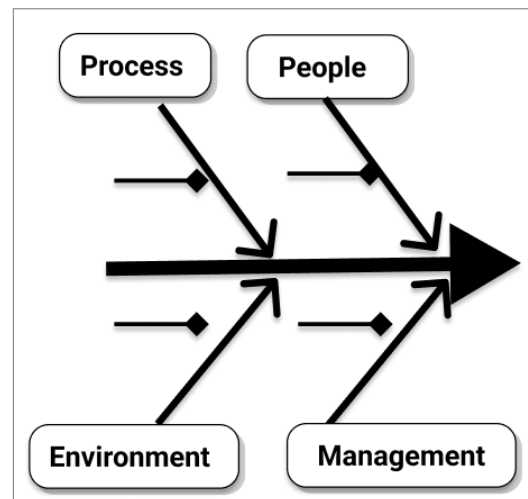
Causal
assumptions

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Causality and relationship

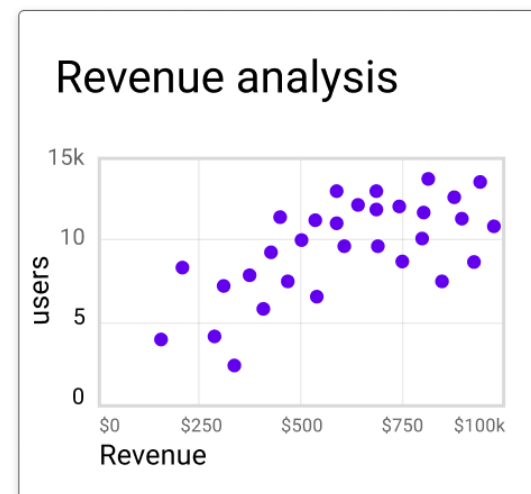
Show causality, mechanism, explanation, systematic structure

Diagram



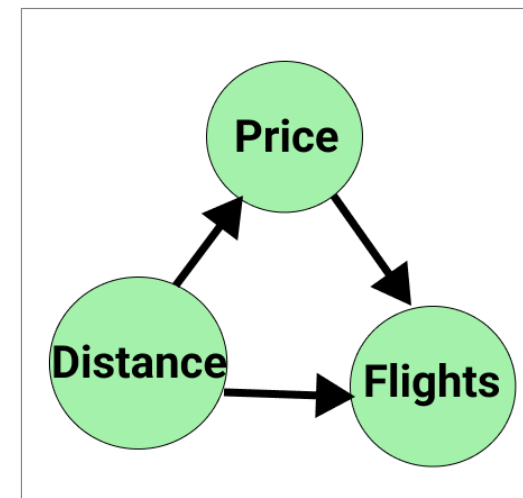
Sorting
ideas

Scatterplot



Simple
relationships

Causal
graph



Causal
assumptions

Flowchart



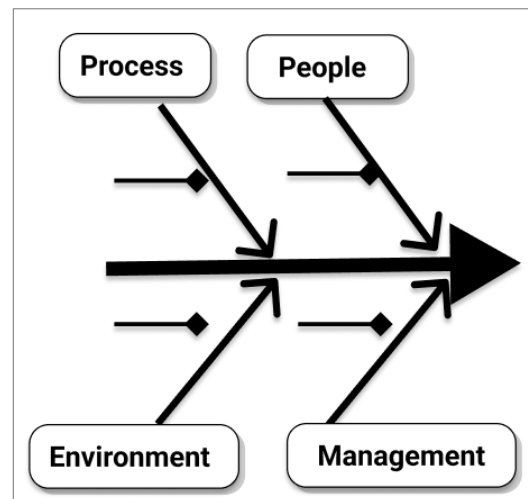
Events or
conditions

¹ Edward Tufte (2006). Beautiful Evidence.

Causality and relationship

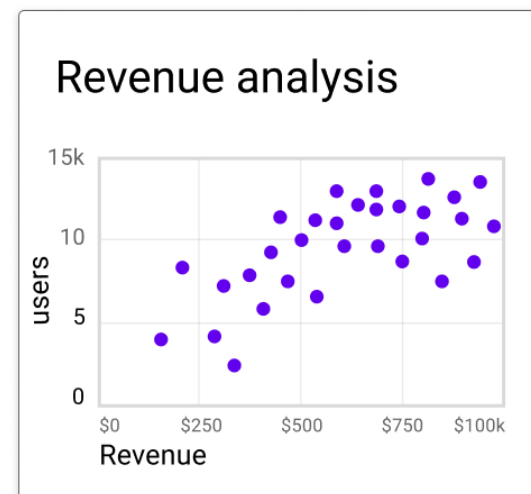
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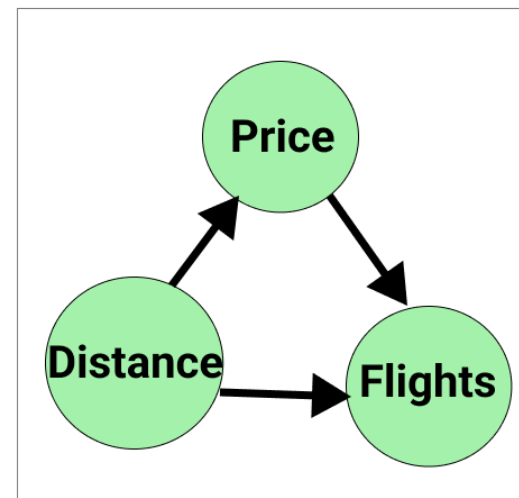
Sorting
ideas

Scatterplot



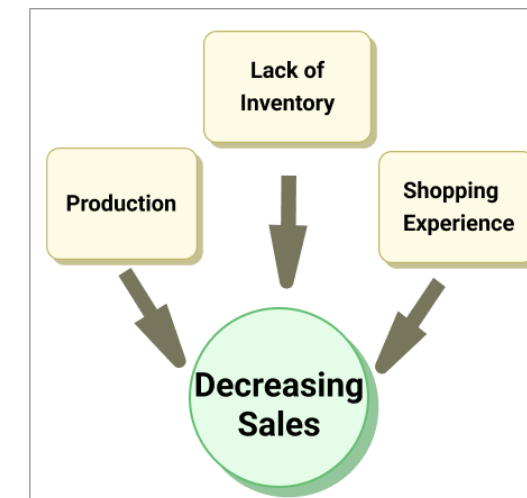
Simple
relationships

Causal
graph



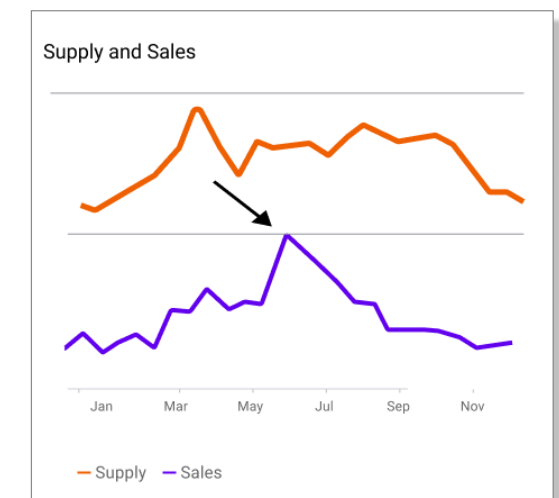
Causal
assumptions

Flowchart



Events or
conditions

Granger-
causality



Probabilistic
causality

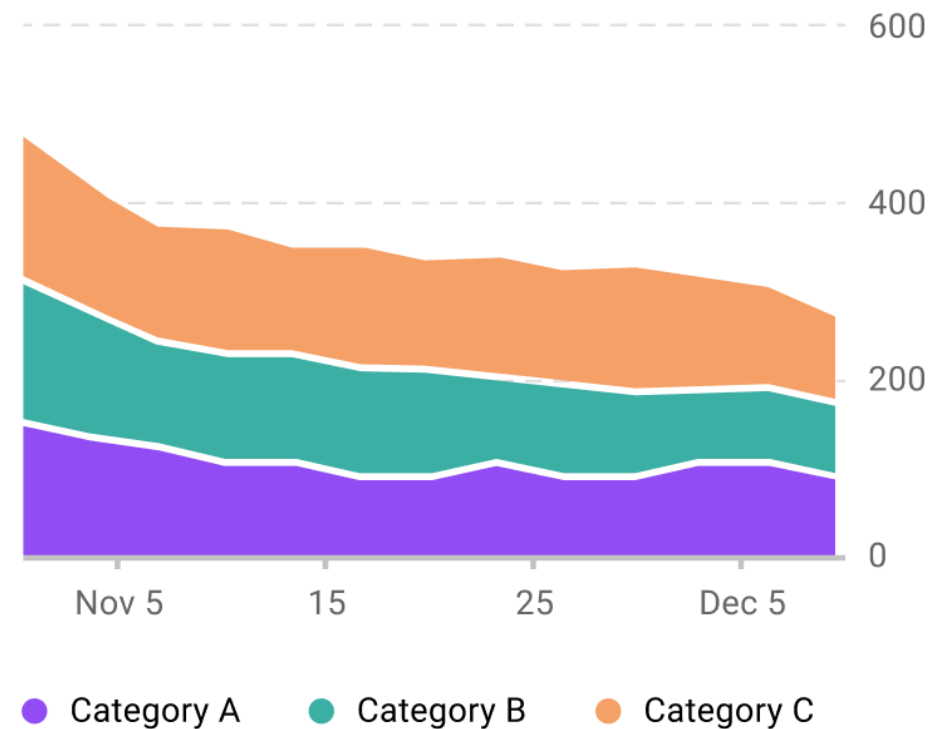
¹ Edward Tufte (2006). Beautiful Evidence.

Multiple variables

Show multiple variables (more than 2 variables and dimensions)

Grouping (single graph)

Users in the last 30 days

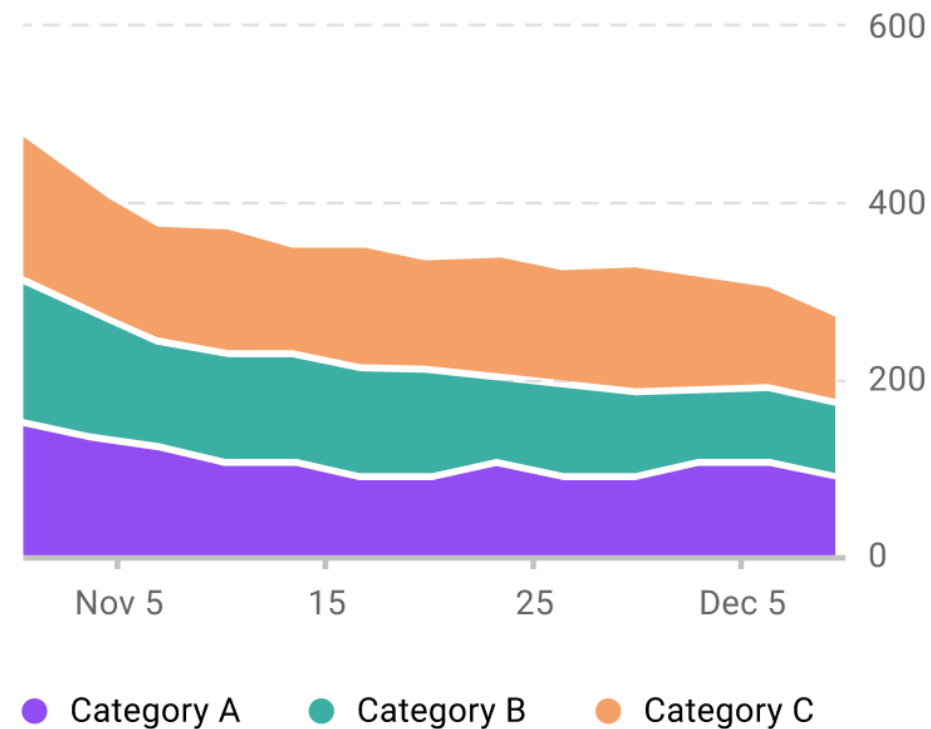


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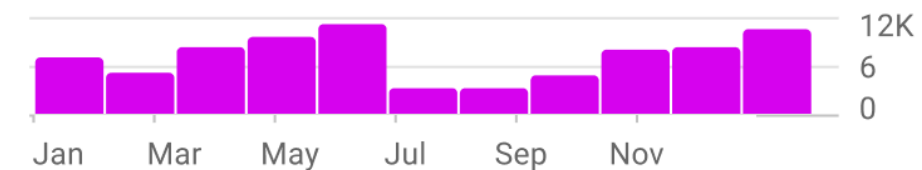
Users in the last 30 days



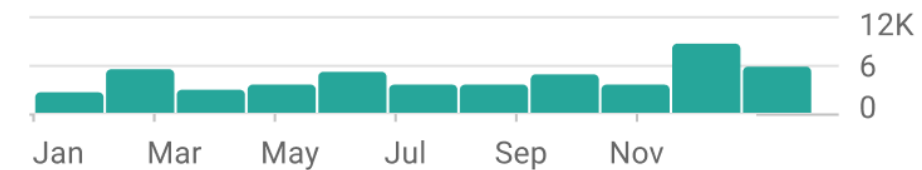
Faceting (multiple graphs)

Monthly totals

Region A



Region B



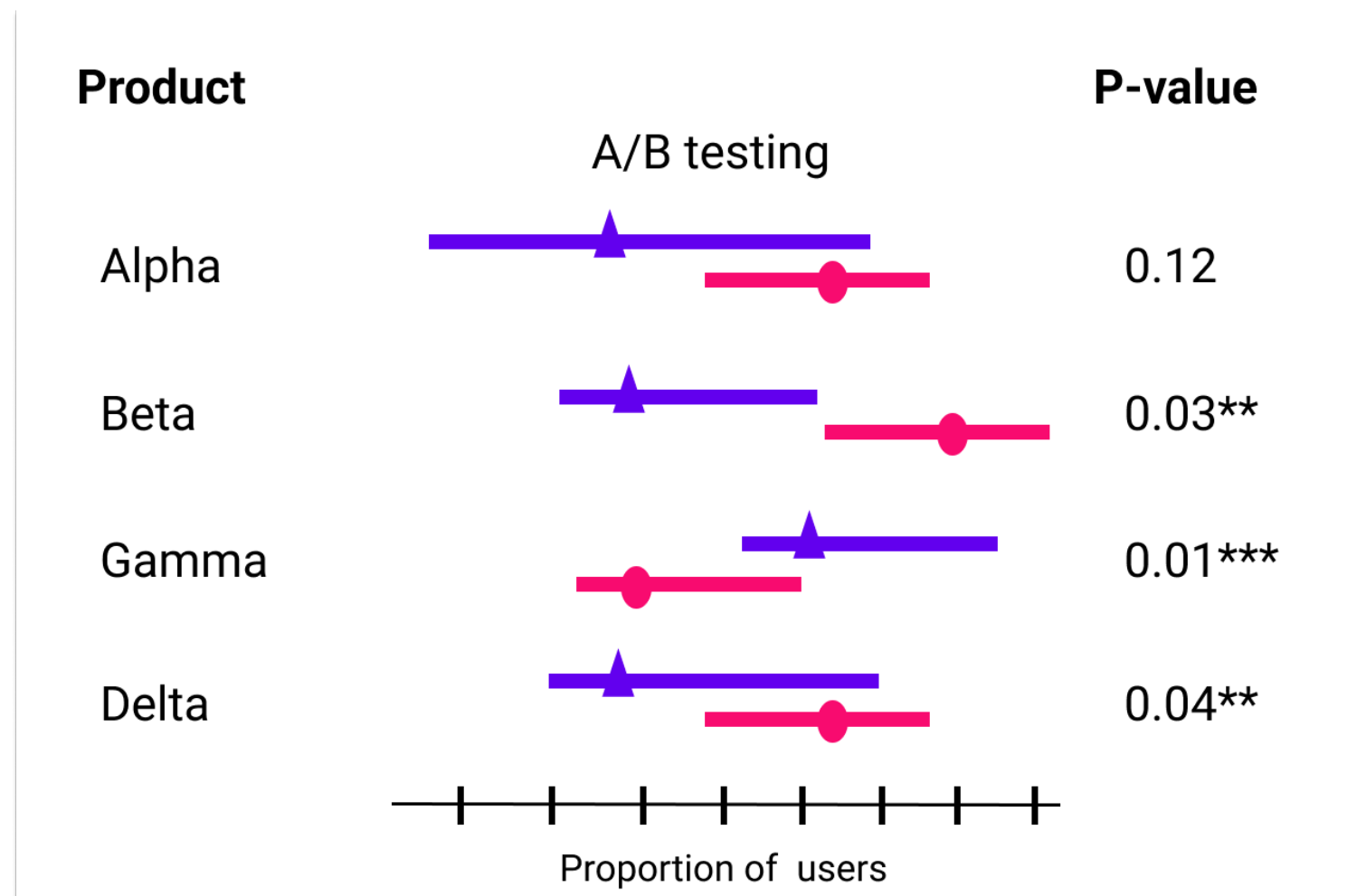
Region C



Integration of evidence

Integrate evidence onto a graphic (text, numbers, images)

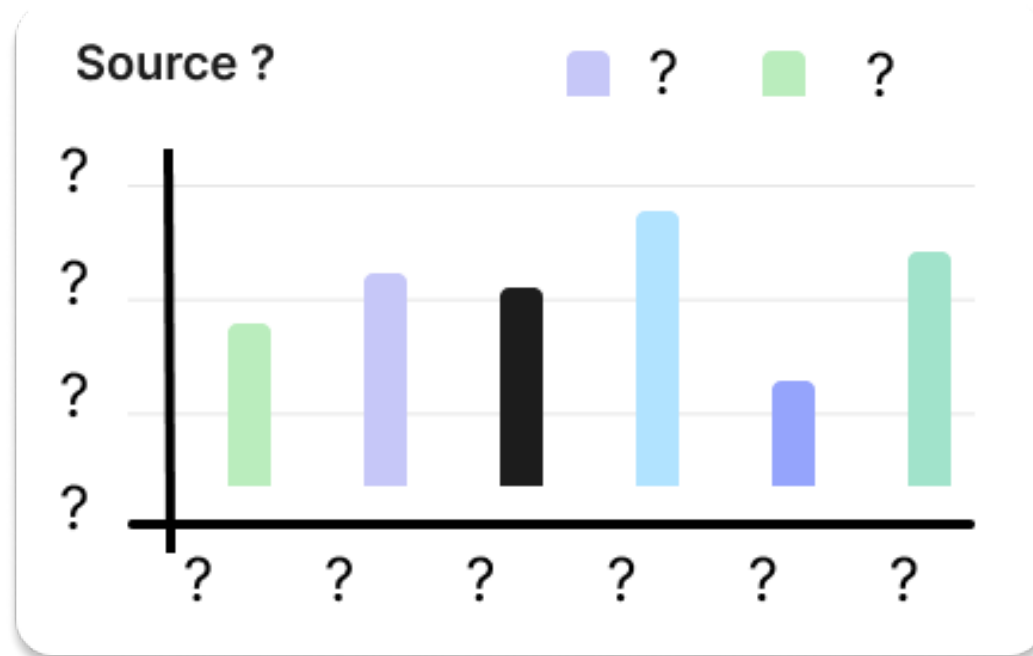
"...what matters entirely is the evidence, not particular modes of evidence"



Documentation and content

Documentation: data, process, description

Content: quality, relevance, and integrity



- What is the data origin?
- How data is processed and cleaned?
- Does the graph have legend and labels?



- Start with a good question



- Develop a sound approach



- Present all the necessary information

Let's practice!
DASHBOARD DESIGN CONCEPTS

Data quality

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD
President, Scrivner Solutions Inc

Top dashboard benefits

Decision-Making

Providing immediate access to
at-a-glance information

¹ Ramos, Diana. (2018). Design a Data Dashboard.

Top dashboard benefits

Decision-Making

Providing immediate access to
at-a-glance information

Time Savings

Tracking multiple sources at once
(time and efficiency)

¹ Ramos, Diana. (2018). Design a Data Dashboard.

Top dashboard benefits

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Transparency

Tracking the changing conditions
of business operations

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Top dashboard benefits

Decision-Making

Providing immediate access to at-a-glance information

Time Savings

Tracking multiple sources at once (time and efficiency)

Transparency

Tracking the changing conditions of business operations

Communication

Encouraging communication between team members

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Top dashboard benefits

Decision-Making

Providing immediate access to at-a-glance information

Time Savings

Tracking multiple sources at once (time and efficiency)

Transparency

Tracking the changing conditions of business operations

Communication

Encouraging communication between team members

Accountability

Increasing accountability among stakeholders

¹ Ramos, Diana. (2018). Design a Data Dashboard.

Common challenges



Irrelevant or
no context



Misleading information



Make data
meaningful

¹ Ramos, Diana. (2018). Design a Data Dashboard.

Common challenges



Irrelevant or
no context



Misleading information



Make data
meaningful



Projected versus
actual



Discrepancies



Understand
your numbers

¹ Ramos, Diana. (2018). Design a Data Dashboard.

Common challenges



Irrelevant or
no context



Misleading information



Make data
meaningful



Projected versus
actual



Discrepancies



Understand
your numbers



Jumping to
conclusion



Wrong causality



Present data
clearly

¹ Ramos, Diana. (2018). Design a Data Dashboard.

Data quality dimensions

Good quality data can power accurate analysis and support trusted business decisions

INTERACTIVE COURSE

Introduction to Data Quality

Completeness

Uniqueness

Integrity

Validity

Accuracy

Consistency

¹ DAMA UK. (2013). Defining data quality dimensions.

Completeness and uniqueness

Completeness: ensures sufficient data

Complete data: no missing entries (null, blank)

Item-level missing data

Person-level missing data

Complete data

Employee NUMBER	VENDOR	BILLING DATE	STATUS	AMOUNT
☐ 5146846548465	✗	2/19/21	Paid	\$500.00
✗	Wade Warren			
☐ 1345705945446	Esther Howard			
☐ 5440754979777	✗	✗	✗	✗
☐ 1243467984543	Brooklyn Simmons	9/18/16		\$500.00
☐ 8454134649707	Leslie Al	5/27/15	Unpaid	\$500.00
☐ 2130164040451	Jenny Wilson	5/27/15	Paid	\$500.00
☐ 0439104645404				

Uniqueness: ensures no duplication

Unique data: item is recorded only once

Top Selling Products

ID	Name	Price	Quantity	Amount
0567	SOS Ridley		82	\$6,518.18
1080	✗ Marco Lightweight Shirt	\$128.50	37	\$4,754.50
1080	✗ Half Sleeve Shirt	\$39.99	64	\$2,559.36
1145	Lightweight Jacket	\$20.00	184	\$3,680.00
4634	Marco Shoes		64	\$1,965.81
5489	OS Ridley High Wa	\$79.49	82	\$6,518.18

Duplicate data

Unique data

Integrity, validity, accuracy, consistency

Integrity

- Ensures that all enterprise data can be traced and connected



Accuracy

- The degree to which data correctly describes the real-world object or event



Validity

- Conforms to the syntax (format, type, range) of its definition



Consistency

- The same information is stored and used at multiple instances



¹ Ashkam et al. (2013). Defining Data Quality Dimensions.

Data quality check

Completeness

How complete is your dataset?

¹ Gupta, Ankur. (2022). The 6 dimensions of data quality.

Data quality check

Completeness

How complete is your dataset?

Accuracy

Is the data an accurate representation of the element it describes?

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Data quality check

Completeness

How complete is your dataset?

Accuracy

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Consistency

Is your data synchronized across your organization?

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Data quality check

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Consistency

Is your data synchronized across your organization?

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Does the data fit the defined range and definition?

¹ Gupta, Ankur. (2022). The 6 dimensions of data quality.

Data quality check

Completeness

How complete is your dataset?

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Are there any duplicates?

¹ Gupta, Ankur. (2022). The 6 dimensions of data quality.

Data quality check

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Are there any duplicates?

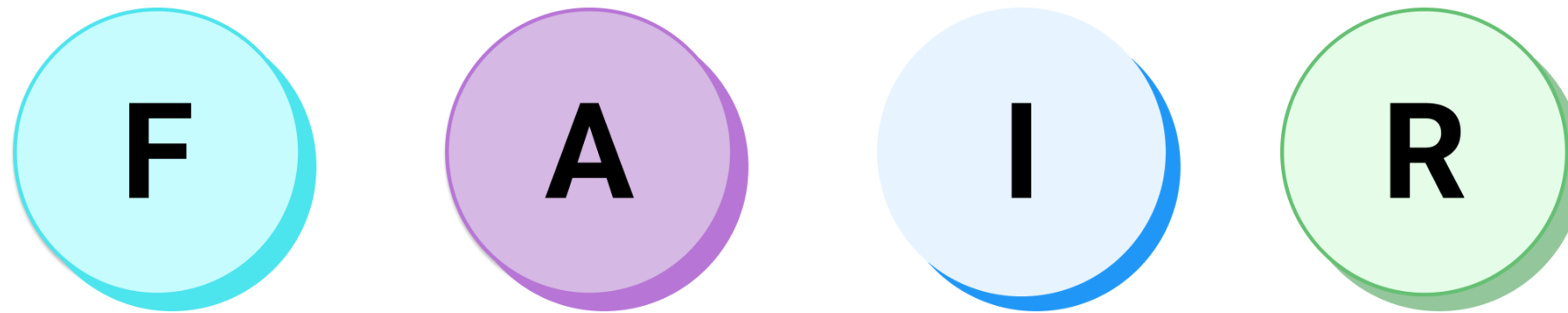
Integrity

Can your data be traced and connected across the organization?

¹ Gupta, Ankur. (2022). The 6 dimensions of data quality.

What is metadata?

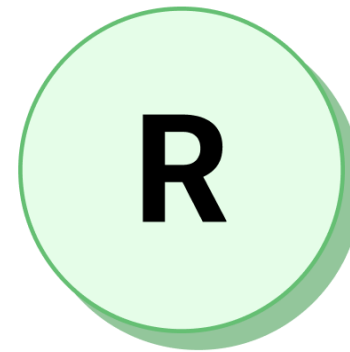
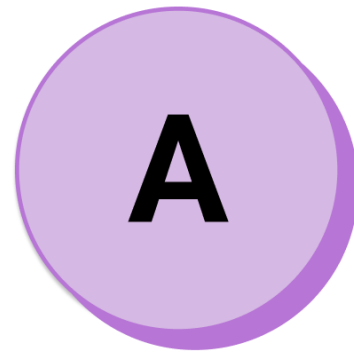
Data about data: when, by whom, how, why



¹ Wilkinson et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship.

What is metadata?

Data about data: when, by whom, how, why



Findable



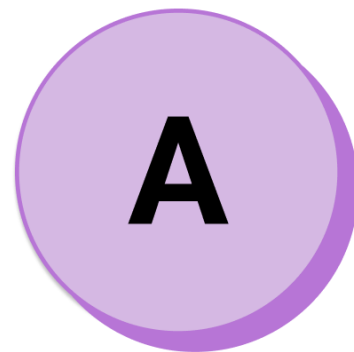
¹ Wilkinson et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship.

What is metadata?

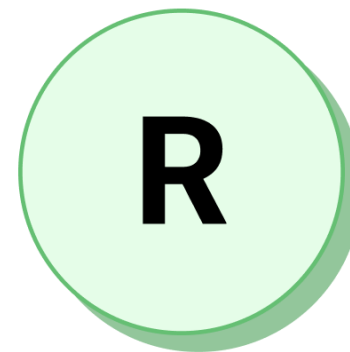
Data about data: when, by whom, how, why



Findable



Accessible



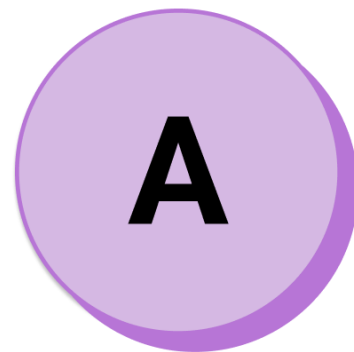
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What is metadata?

Data about data: when, by whom, how, why



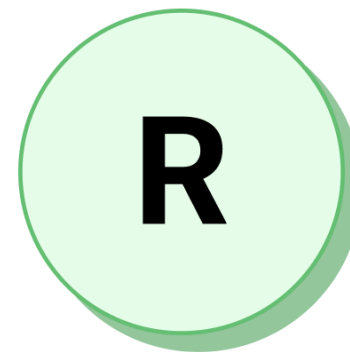
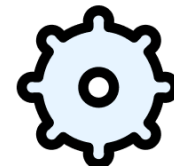
Findable



Accessible



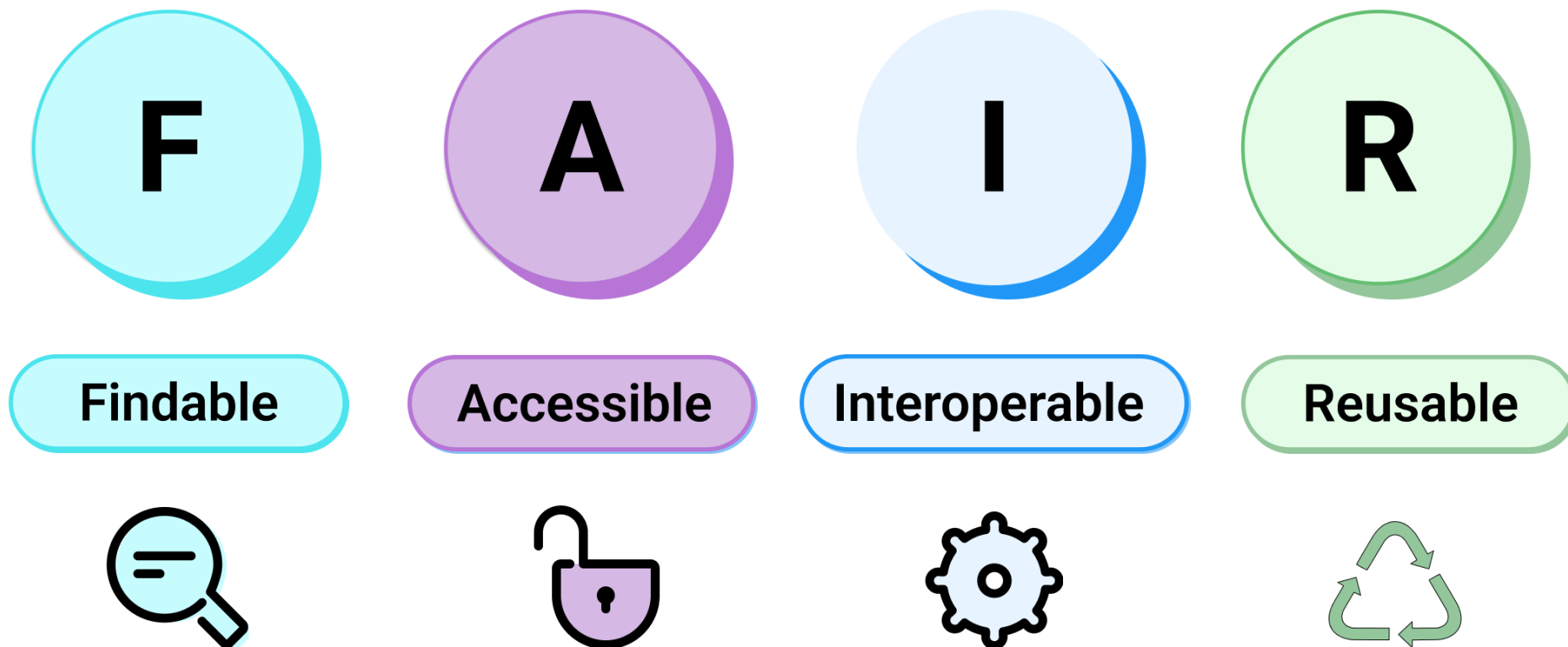
Interoperable



¹ Wilkinson et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship.

What is metadata?

Data about data: when, by whom, how, why



¹ Wilkinson et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship.

Metadata and dashboard

Data reliability: trusted sources

Data selection: appropriate data gathering

Data reproducibility: ability to get similar results when applying the same design

Data imputation: appropriate reduction of anomalies caused by missing data



Let's practice!
DASHBOARD DESIGN CONCEPTS